

COURSE MATERIAL

SUBJECT	PART-B BASIC ELECTRONICS ENGINEERING (EE23AES101)
UNIT	1
COURSE	B.TECH
DEPARTMENT	ECE
SEMESTER	1-1
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1. Course Objectives

The objectives of this course are

1. To understand the working principle of various electronic devices.
2. To acquire knowledge on the operating conditions of a DC power supply system.
3. To learn basic techniques in the design of a logic circuit.

2. Prerequisites

Students should have knowledge on

1. Applied Physics
2. Network Analysis

3. Syllabus

UNIT I

Introduction - Evolution of electronics - Characteristics of PN Junction Diode - Zener Effect- Zener Diode and its Characteristics. Bipolar Junction Transistor - CB, CE, CC Configurations and Characteristics - Elementary Treatment of Small Signal CE Amplifier.

4. Course outcomes

After completion of this subject, students will be able to

1. **Apply the concept of science and mathematics to understand the working principles of electronic devices.**
2. **Analyze the working principle of a DC power supply system and Amplifiers.**
3. **Solve digital logic circuits and implement using different logic gates.**

5. Co-PO / PSO Mapping

EDC	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	3	2	1										3	2
CO2	3	3	3	2									3	3
CO3	3	3	2	2									3	3

6. Lesson Plan

Lecture No.	Weeks	Topics to be covered	References
1	1	Introduction - Evolution of electronics	T1
2		Characteristics of PN Junction Diode	T1
3		Zener Effect	T1
4		Zener Diode and its Characteristics	T1
5	2	Bipolar Junction Transistor	T1
6		Bipolar Junction Transistor - CB, CE, CC Configurations and Characteristics	T1
7		Elementary Treatment of Small Signal CE Amplifier.	T1

7. Activity Based Learning

1. Explanation of operation of Intrinsic and Extrinsic semiconductors.
2. Application of diodes through simulation tools.

8. Lecture Notes

1.1 Introduction- Evolution of electronics

Electronics means study of flow of electrons in electrical circuits. The word Electronics comes from electron mechanics which means learning the way how an electron behaves under different conditions of externally applied fields. IRE - The Institution of Radio Engineers has given a definition of electronics as "that field of science and engineering, which deals with electron devices and their utilization." Fundamentals of electronics are the core subject in all branches of engineering nowadays.

WHAT IS ELECTRONICS?

The world's reliance on electronics is so great that commentators claim people live in an "electronic age." People are surrounded by electronics—televisions, radios, computers, mobiles, Laptop and DVD players, along with products with major electric components, such as microwave ovens, refrigerators, and other kitchen appliances, automatic vehicles, Robotics, as well as hearing aids and medical instruments and numerous applications in industry.

Definition: The branch of engineering which deals with current conduction through a Vacuum or Gas or Semiconductor is known as Electronics. An electronic device is that in which current flows through a vacuum or gas or semiconductor. This control of electrons is accomplished by devices that resist, carry, select, steer, switch, store, manipulate, and exploit the electron.

(Or)

Electronics deals with electrical circuits that involve active electrical components such as vacuum tubes, transistors, diodes and integrated circuits, and associated passive interconnection technologies. Commonly, electronic devices contain circuitry consisting primarily or exclusively of active semiconductors supplemented with passive elements; such a circuit is described as an electronic circuit.

HISTORY OF ELECTRONICS

Invention of Vacuum Tubes

Electronics took birth in 1897 when J.A. Fleming developed a vacuum diode. Useful electronics came in 1906 when vacuum triode was invented by Lee De Forest. This device could amplify electrical signals. Later, around 1925, tetrode and pentode vacuum tubes were developed. These tubes dominated the field of electronics till the end of World War II.

Invention of Transistor

The era of semiconductor electronics began with the invention of the junction transistor in 1948 at Bell Laboratories. Soon, the transistors replaced the bulky vacuum tubes in different electronic circuits. The tubes had major limitations: power was consumed even when they were not in use and filaments burnt out.

Earlier, the transistors were made of germanium, as it was easier to purify. In 1954, silicon transistors were developed. These afforded operations up to 200°C, whereas germanium device could work well only up to about 75°C. Today, almost all semiconductor devices are fabricated using silicon.

Invention of the Integrated Circuits (IC)

In 1958, Jack Kilby conceived the concept of building an entire electronic circuit on a single semiconductor chip. Later, all active and passive components and their interconnections could be integrated on a single chip. This drastically reduced the size and weight, as well as the cost of electronic equipment.

The following approximate data give some indication of the increasing component count per chip of area $3 \times 5 \text{ mm}^2$ and thickness comparable with human hair.

1951 - Discrete transistors

1960 - Small-Scale Integration (SSI), fewer than 100 transistors.

1966 - Medium-Scale Integration (MSI), 100 to 1000 transistors.

1969 - Large-Scale Integration (LSI), 1000 to 10000 transistors.

1975 - Very-Large-Scale Integration (VLSI), more than 10000 transistors.

1994 - Ultra-large-scale integration (ULSI) more than 1 million transistors.

2012 - INTEL introduced a computer processor chip (62-Core Xeon Phi) containing 5,000,000,000 transistors.

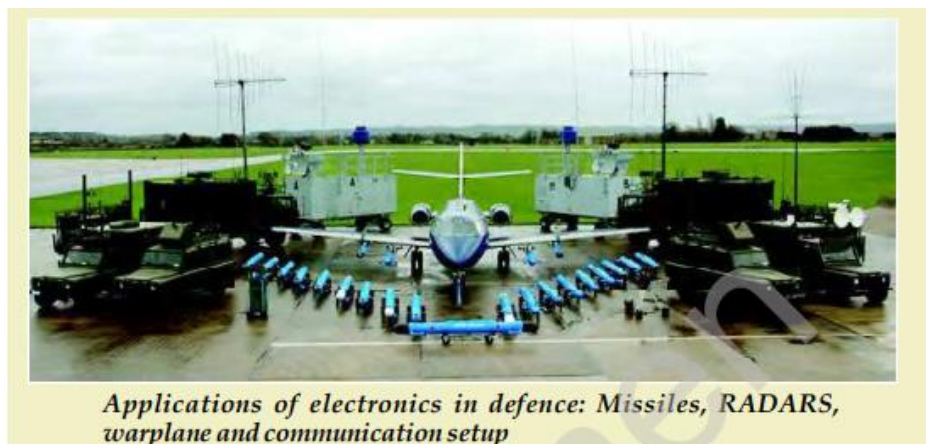
APPLICATIONS OF ELECTRONICS

Electronics plays an important role in almost every sphere of our life. Electronics has penetrated in every field from an ordinary wrist watch to super computers; from telephone repeaters specimen 11 buried deep under sea to the satellites far out in space; from the control of modern household appliances to the control of super tankers carrying cargo across the sea.

Communication and Entertainment: The progress of a nation depends upon the availability of cheaper and faster means of communication. The main application of electronics in the beginning was in the field of telephony and telegraphy. These utilize a pair of wires as communication channel. Later it was possible to transmit any message from one place to another without wires (wireless communication). Satellite communication has reduced the distance between people and places.

Radio and TV broadcasting provide a means of both communication as well as entertainment. Electronic gadgets like tape recorders, music and video players, stereo systems, public address systems, etc. are widely used for entertainment.

Applications in Defense sector: In a war, success or defeat of a nation depends on the reliability of its communication system. In modern warfare, communication is almost entirely electronic. Guided missiles are completely controlled by electronic circuits. One of the most important developments during World War II was the RADAR (Radio Amplification Detection And Ranging). By using RADAR it is possible not only to detect, but also to find the exact location of the enemy aircraft. The anti-aircraft guns can then be accurately directed to shoot down the aircraft. In fact the RADAR and anti-aircraft guns can be linked by an automatic control system to make a complete unit.



Instrumentation: Instrumentation plays a very important role in any industry and research organization, for precise measurement of various quantities. Very accurate and user-friendly instruments like digital voltmeter (DVM), cathode ray oscilloscope (CRO), frequency counter, signal generator, strain gauge, pH-meter, spectrum analyzers, etc. are some of the electronic equipment without which no research laboratory is complete.

Medical Electronics: Electronic equipment are being used extensively in medical field. They not only assist in diagnosis but also help in the researches that provide treatment and cure for illnesses and even genetic anomalies. Examples are Electron microscope, ECG, EEG, X -rays, defibrillator, oscilloscopes, MRI, CT scanner, glucometer, etc. Some of the instruments are shown in the fig.



Fig. Some applications of electronics in medical field

Applications in Industries: Use of automatic control systems in different industries is increasing day by day. The thickness, quality and weight of a material can be easily controlled by electronic circuits. Electronic circuits are used to control the operations of automatic door openers, lighting systems, power systems, safety devices, etc. Use of computer has made the ticket reservations in railways and airways simple and convenient. Even the power stations, which generate thousands of megawatts of electricity, are controlled by electronic circuits.

Applications in Automobiles: Several electronic equipment are used in cars for charging battery, enabling power assisting functions, measuring gauges and monitoring and controlling the engine performance. The most important application is electronic ignition, which provides better timing of the ignition spark, especially at high speeds. Automobile industry is one of the fastest growing sectors in the world. The end users are demanding greater fuel efficiency, security and safety. This is possible because of the rapid development in the technology. Other areas of application in automobile are parking sensors, auto wipers, auto lights, safety (eg. Air bags), security, anti-

theft systems, etc.

Consumer Electronics: We use fans in our home, class rooms, library, etc. You are familiar with the electronic regulators used with them. Have you ever thought of the mechanism behind that? Here we use an electronic component known as TRIAC to control the speed of the fan. The speed of the fan is directly proportional to the electric power reaching the motor. The regulator controls the speed by controlling the electric power. The regulator controls electric power according to the position of the knob. Special electronic components like Silicon Controlled Rectifiers (SCRs) are used in the speed-control of motors, power rectifiers and inverters.

Home appliances are used 24 hours a day, 7 days a week. It includes personal computers, telephones, audio equipment, televisions, calculators, washing machines, DVD players, etc. Some home appliances are shown in the fig.



Fig. (a) Washing machine (b) TV (c) Radio (d) Mobile Phone

ELECTRONIC COMPONENTS - ACTIVE AND PASSIVE Electronic components can be broadly classified into active and passive components. Active Components are electronic components which are capable of amplifying or processing an electrical signal, e.g.: Diodes, Transistors, etc. Passive Components are electronic components which are not capable of amplifying or processing an electrical signal, e.g.: resistors, capacitors and inductors.

RESISTORS: A resistor is a two terminal component which provides resistance to the flow of current in a circuit. The symbols are shown in fig. 1.4. All resistors have power ratings. It is the maximum power that can be dissipated without damaging the component. Thus, a 1 watt resistor with a resistance of 100 can pass a maximum current of 100 mA. The size of a resistor is usually bigger if its wattage rating is higher, so as to withstand higher power dissipation. Resistors can be broadly classified into two groups-fixed and variable.

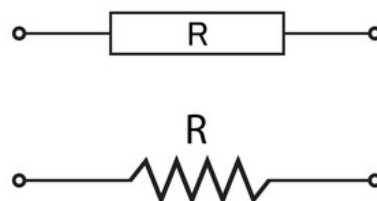


Fig. Symbols of Resistor

Fixed Resistors: A fixed resistor is one for which the value of its resistance is specified and cannot be varied in general. These resistors may be carbon-composition resistors, carbon film resistors or wire wound resistors.

Colour coding of Resistors: Carbon-composition and carbon film resistors are small in size. It becomes almost impossible to print the ratings on their body. Therefore, a standard colour coding is used to indicate the ratings. The resistance is given in the form of four (or five) coloured signs (or bands) painted on the body. The coloured bands are always read from left to right from the end that has the bands closer to it, as shown in Fig.

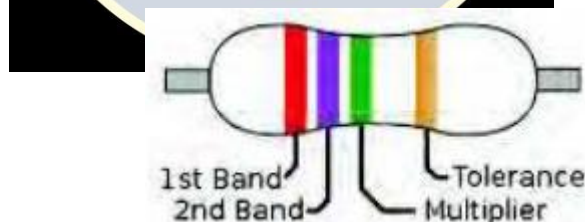


Fig. Colour coding of Resistors

The first and second colour bands represent the first and second numbers (significant digits) of the resistance value. The third band indicates the multiplication factor. The fourth band represents tolerance. It is a measure of the precision with which the resistor was manufactured. In case the fourth band is not present, the tolerance is assumed $\pm 20\%$. Resistance value calculation from colour coding is shown in Fig.

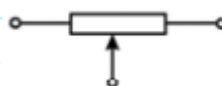
Color	Digit value	Multiplier (No. of zeros)	Tolerance
Black	0	10^0	-
Brown	1	10^1	$\pm 1\%$
Red	2	10^2	$\pm 2\%$
Orange	3	10^3	-
Yellow	4	10^4	-
Green	5	10^5	$\pm 0.5\%$
Blue	6	10^6	$\pm 0.25\%$
Violet	7	10^7	$\pm 0.1\%$
Gray	8	10^8	$\pm 0.05\%$
White	9	10^9	-
Gold			$\pm 5\%$
Silver			$\pm 10\%$
No Color	-	-	$\pm 20\%$

Fig. Colour Coding Table

For example a resistor has a colour band sequence: yellow, violet, orange and gold. With the help of the colour coding table in Fig. 1.9, we can calculate the value of this resistor as shown in the table below.

1 st band	2 nd band	3 rd band	4 th band	Value
Yellow	Violet	Orange	Gold	
4	7	10^3	$\pm 5\%$	$47k\ \Omega \pm 5\%$

Variable Resistors: In electronic circuits, sometimes it becomes necessary to adjust the values of currents and voltages while the equipment is in use. For example, it is often desired to change the volume of sound, the brightness of a television picture, etc. Variable resistors can be used to do such adjustments. Big size variable resistors are usually called rheostats. In electronic circuits, we use small size variable resistors, and they are called potentiometers (usually abbreviated to 'pots'). The symbol for potentiometer is given in Fig.



(a) Symbol for potentiometer

Ohm's law states the relationship between electric current and potential difference. The current that flows through most conductors is directly proportional to the voltage applied to it. Mathematically, this current-voltage relationship is written as, $V=IR$.

The same formula can be rewritten in order to calculate the current and resistance respectively as follows: $I=V/R$ and $R=V/I$

CAPACITORS: A capacitor can store electrical energy in its electric field, and release it whenever desired. A capacitor opposes any change in the potential difference applied across its terminals. The capacitance of a capacitor is measured in farads (F). Capacitors vary in shape and size, but the basic configuration is two conductors separated by an insulating medium. The simplest example of a capacitor consists of two conducting plates of area A , which are parallel to each other, and separated by a distance d , as shown in Figure.

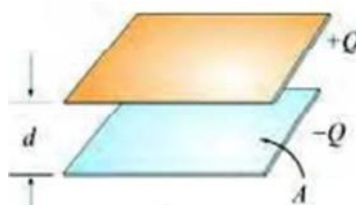


Fig. Parallel-plate capacitor

The capacitance of any parallel-plate capacitor can be calculated by the formula; $C= \epsilon A/d$

Where C is capacitance in farads and ϵ is permittivity of the dielectric.

the amount of charge Q stored in a capacitor is directly proportional to the electric potential difference (V) between the plates i.e., $Q = C V$

Similar to resistor, a capacitor offers resistance to a signal passing through it. This is known as capacitive reactance. Capacitive reactance is given as:

$$X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

Where X_c is the capacitive reactance, ω is the angular frequency, f is the frequency in Hertz, and C is the capacitance. The unit of capacitive reactance is ohm. It is clear from the above equation that the reactance of a capacitor increases with decreasing frequency.

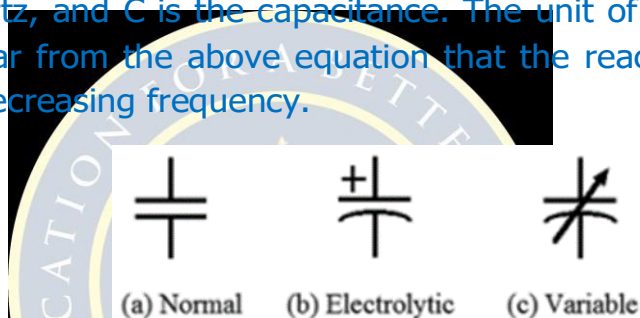


Fig. Capacitor symbols

Like resistors, capacitors can either be fixed or variable. Some of the most commonly used fixed capacitors are mica, ceramic, paper, and electrolytic. Variable capacitors are mostly air-gang capacitors.



Fig. mica capacitor, Ceramic disk capacitor, Electrolytic capacitor and Air-gang capacitor (variable).

INDUCTORS: An inductor is a device that stores electrical energy in the magnetic field surrounding it. Inductance is the property of a coil (the inductor) to oppose a change in current. In its simplest form, an inductor consists of a wire loop or coil. The inductance of an inductor is directly proportional to the number of turns in the coil. Inductance also depends on the radius of the coil and on the type of material (core) around which the coil is wound. The schematic symbols for inductors are shown in figure 1.21. The unit of Inductance is henry (H). An inductor offers opposition to the passage of any change in current through it. This opposition is called inductive reactance. Inductive reactance is defined as

$$X_L = 2\pi f L = \omega L.$$

where X_L is the inductive reactance, ω is the angular frequency, f is the

frequency in Hertz, and L is the inductance. The unit of inductive reactance is Ohm. The reactance of an inductor increases with increase in frequency.

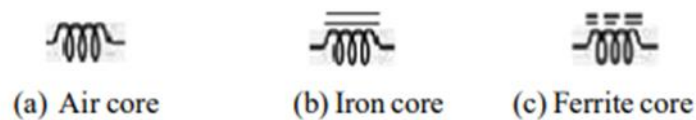
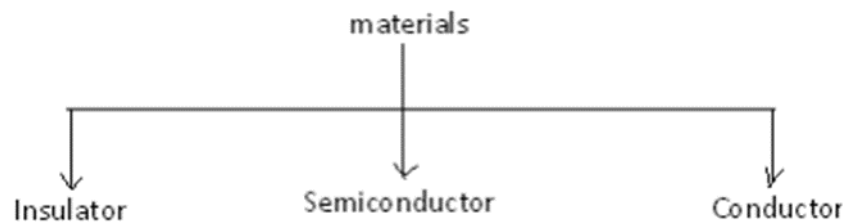


Fig. Symbols of Inductors

1.2 Characteristics of PN Junction Diode

Review of Semiconductors: Based on the electrical conductivity all the materials in nature are classified as insulators, semiconductors, and conductors.



Insulator: An insulator is a material that offers a very low level (or negligible) of conductivity when voltage is applied. Eg: Paper, Mica, glass, quartz. Typical resistivity level of an insulator is of the order of 10^{10} to $10^{12} \Omega\text{-cm}$. The energy band structure of an insulator is shown in the fig. The energy band between the valance band and conduction band is called as forbidden band gap. It is the energy required by an electron to move from valance band to conduction band i.e. the energy required for a valance electron to become a free electron. $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$. For an insulator, as shown in the fig.1.1 there is a large forbidden band gap of greater than 5 eV . Because of this large gap there are very few electrons in the CB and hence the conductivity of insulator is poor. Even an increase in temperature or applied electric field is insufficient to transfer electrons from VB to CB.

Conductors: A conductor is a material which supports a generous flow of charge when a voltage is applied across its terminals. i.e. it has very high conductivity. Eg: Copper, Aluminium, Silver, Gold. The resistivity of a conductor is in the order of 10^{-4} and $10^{-6} \Omega\text{-cm}$. The Valance and conduction bands overlap (fig1.1) and there is no energy gap for the electrons to move from valance band to conduction band. This implies that there are free electrons in CB even at absolute zero temperature (0K). Therefore, at room temperature when electric field is applied large current flows through the conductor.

Semiconductor: A semiconductor is a material that has its conductivity somewhere between the insulator and conductor. The resistivity level is in the range of 10 and $10^4 \Omega\text{-cm}$. Two of the most commonly used are Silicon ($\text{Si}=14$ atomic no.) and germanium ($\text{Ge}=32$ atomic no.). Both have 4 valance electrons. The forbidden band gap is in the order of 1 eV . For eg., the band gap energy for Si, Ge and GaAs is 1.21 , 0.785 and 1.42 eV , respectively at absolute zero temperature (0K). At 0K and at low temperatures, the valance band electrons do not have sufficient energy to move from V to CB. Thus semiconductors act as insulators at 0K. At room temperature there are sufficient electrons in the CB and hence the semiconductor is capable of conducting some current at room temperature.

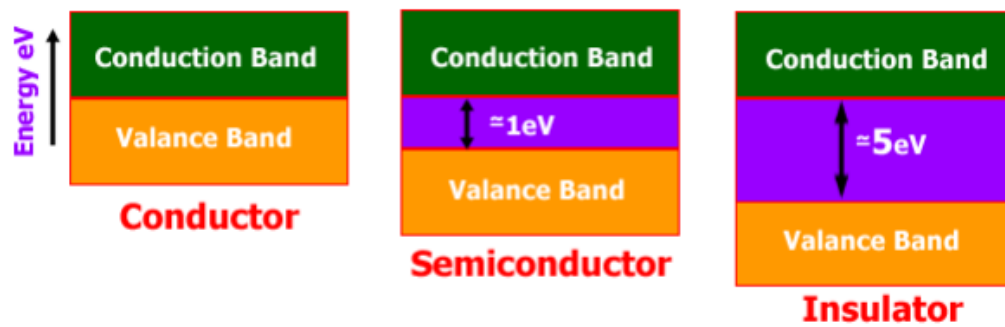
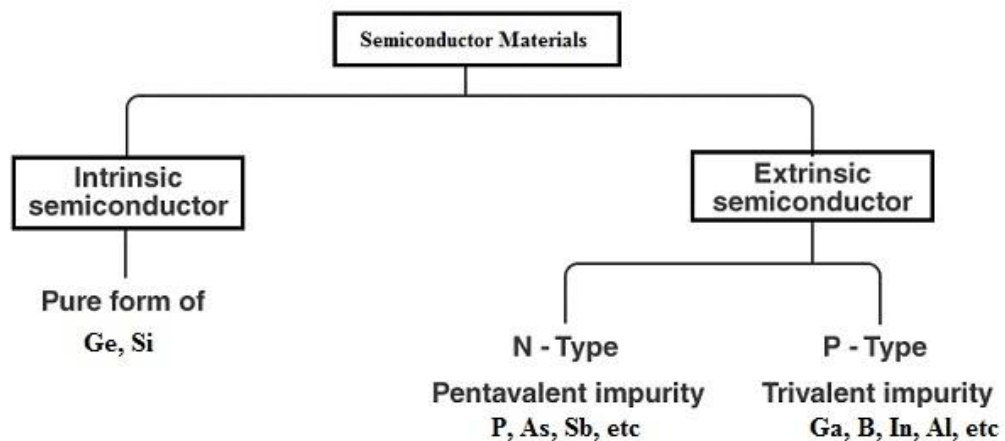


Fig: Energy band diagrams insulator, semiconductor and conductor

SEMICONDUCTOR TYPES:



A pure form of semiconductors is called as intrinsic semiconductor. In a pure semiconductor, the number of holes is equal to the number of free electrons. Conduction mechanism of intrinsic semiconductor is illustrated in the below figure.

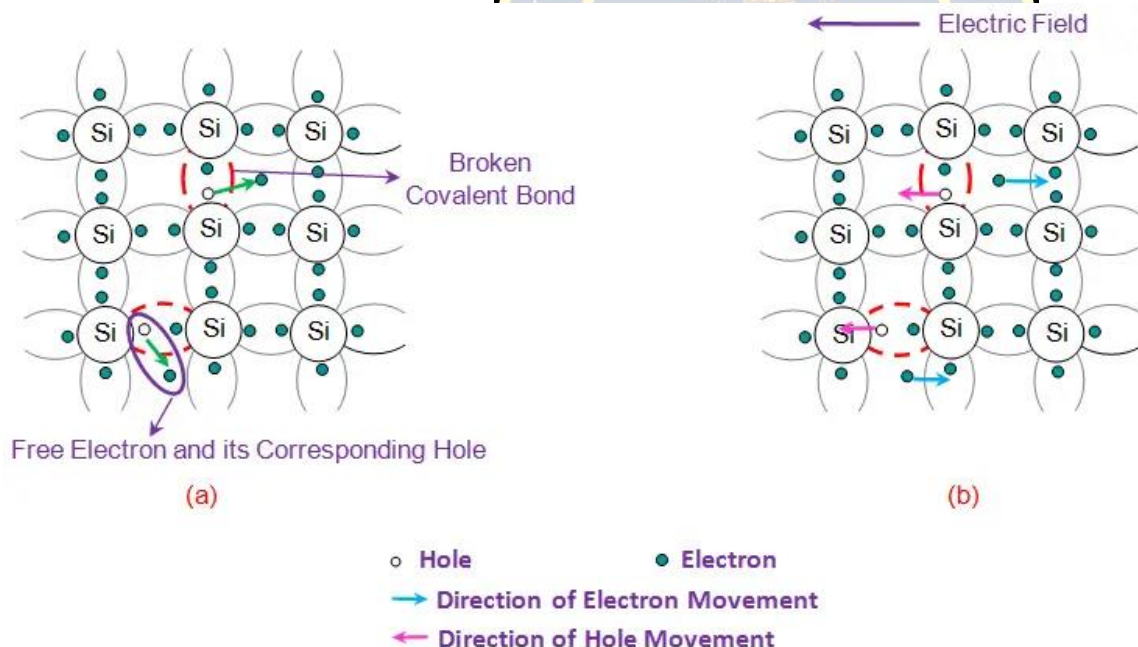


Figure 3 Conduction Mechanism in case of Intrinsic Semiconductors in the
(a) Absence of Electric Field (b) Presence of Electric Field

INTRINSIC SEMICONDUCTOR: Intrinsic semiconductor has very limited applications as they conduct very small amounts of current at room temperature. The current conduction capability of intrinsic semiconductor can be increased

significantly by adding a small amounts impurity to the intrinsic semiconductor. By adding impurities it becomes impure or extrinsic semiconductor. This process of adding impurities is called as doping. The amount of impurity added is 1 part in 10⁶ atoms.

N type semiconductor: If the added impurity is a pentavalent atom then the resultant semiconductor is called N-type semiconductor. Examples of pentavalent impurities are Phosphorus, Arsenic, Bismuth, Antimony etc. A pentavalent impurity has five valance electrons. Fig. a shows the crystal structure of N-type semiconductor material where four out of five valance electrons of the impurity atom(antimony) forms covalent bond with the four intrinsic semiconductor atoms. The fifth electron is loosely bound to the impurity atom.

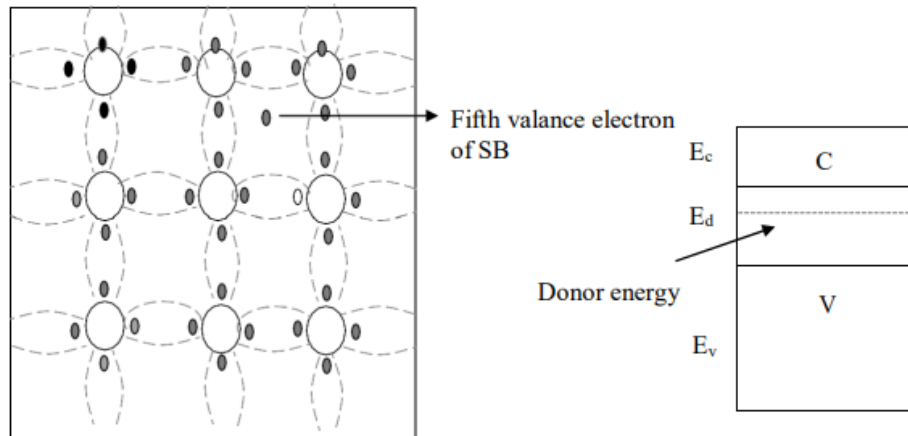


Fig. a crystal structure of N type

Fig. b Energy band diagram of N type

In the N-type semiconductor (sc) the no. of electrons increases and the no. of holes decreases compared to those available in an intrinsic sc. The reason for decrease in the no. of holes is that the larger no. of electrons present increases the recombination of electrons with holes. Thus, current in N type sc is dominated by electrons which are referred to as majority carriers. Holes are the minority carriers in N type sc.

P type semiconductor: If the added impurity is a trivalent atom, then the resultant semiconductor is called P-type semiconductor. Examples of trivalent impurities are Boron, Gallium, indium etc. The crystal structure of p type semiconductor (sc) is shown in the fig. c. The three valance electrons of the impurity (boon) forms three covalent bonds with the neighbouring atoms and a vacancy exists in the fourth bond giving rise to the holes. The hole is ready to accept an electron from the neighbouring atoms. Each trivalent atom contributes to one hole generation and thus introduces a large no. of holes in the valance band. At the same time the no. electrons are decreased compared to those available in intrinsic semiconductor (sc) because of increased recombination due to creation of additional holes.

Thus, in P type sc, holes are majority carriers and electrons are minority carriers. Since each trivalent impurity atoms are capable accepting an electron, these are called as acceptor atoms. The following fig . d shows the pictorial representation of P type semiconductor.

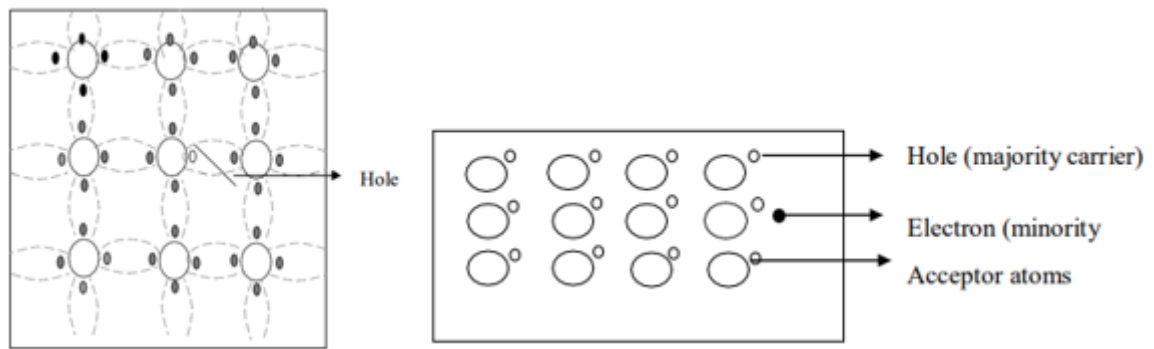
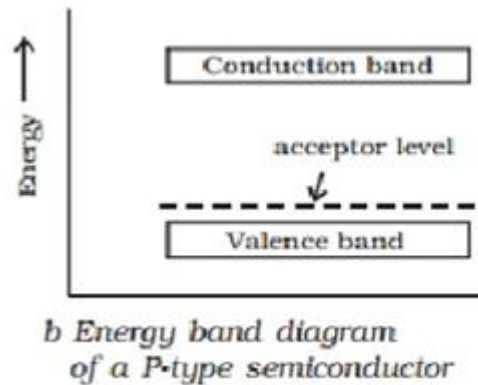


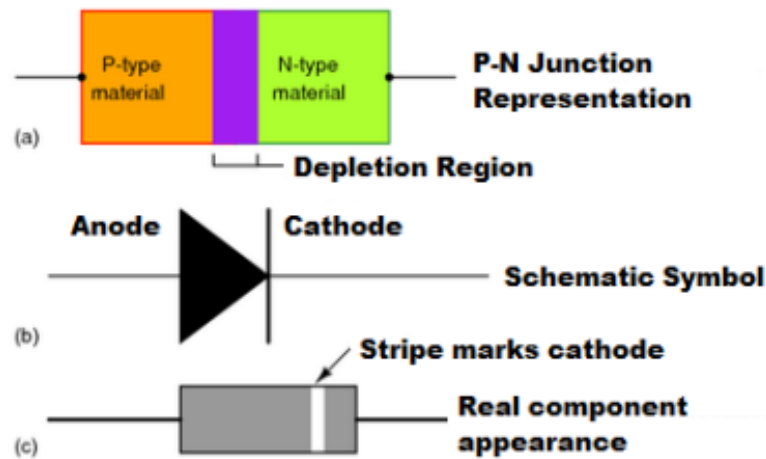
Fig. crystal structure of P type



The conductivity of N type sc is greater than that of P type sc as the mobility of electron is greater than that of hole.

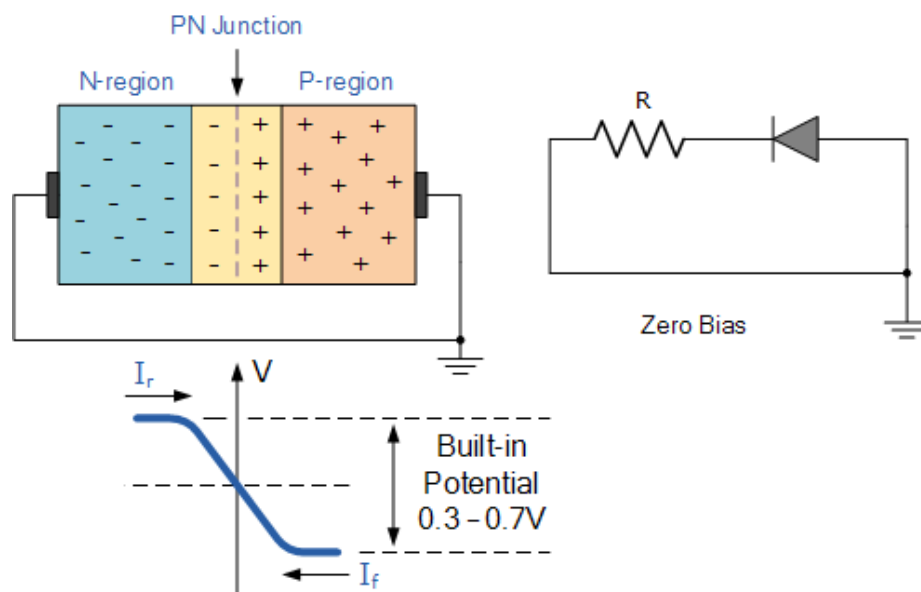
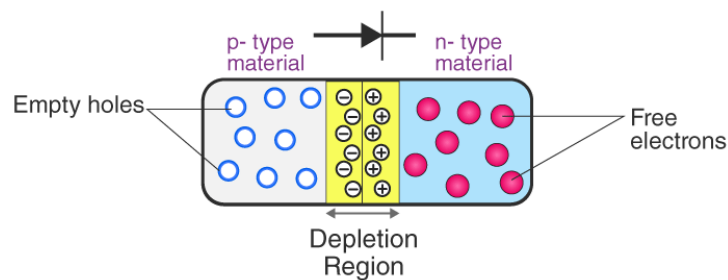
PN JUNCTION DIODE:

- A p-n junction diode is a two-terminal or two-electrode semiconductor device, which allows the electric current in only one direction while blocks the electric current in opposite or reverse direction.
- A diode is defined as a two-terminal electronic component that only conducts current in one direction. Ideal diode will have zero resistance in one direction, and infinite resistance in the reverse direction.
- In n-type semiconductors, free electrons are the majority charge carriers whereas in p-type semiconductors, holes are the majority charge carriers. When the n-type semiconductor is joined with the p-type semiconductor, a p-n junction is formed.
- The p-n junction, which is formed when the p-type and n-type semiconductors are joined, is called as p-n junction diode. The p-n junction diode is made from the semiconductor materials such as silicon, germanium, and gallium arsenide. The basic symbol of p-n junction diode
- A PN junction diode is also known as a semiconductor diode. It is formed when a Ptype and an N-type semiconductor are joined metallurgically. It allows current flow in one direction only.
- Hence, it can act as rectifier to convert ac voltage to dc voltage.
- The arrowhead indicates the conventional direction of current flow when the diode is forward biased.
- The P-side of the diode is positive and is called anode. The N-side is the cathode and is the negative terminal when the diode is forward biased.
- The word 'diode' stands for two electrodes- anode and cathode.



UNBIASED or ZERO Bias PN-JUNCTION Diode: When a diode is Zero Biased no external energy source is applied and a natural Potential Barrier is developed across a depletion layer, which is approximately 0.5 to 0.7v for silicon diodes and approximately 0.3 of a volt for germanium diodes. The junction region itself has no charge carriers and is known as the depletion region. The junction (depletion) region has a physical thickness that varies with the applied voltage.

UNBIASED P-N JUNCTION

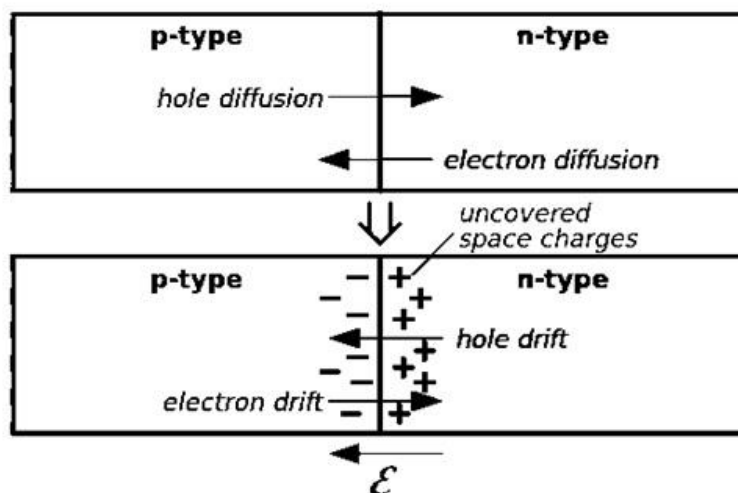


- The potential difference required for the electrons to be passed across the electric field is called the potential barrier.
- There is a very thin region called the depletion region in the middle. In this region, all the holes and the electrons from both sides combine together, and hence only ions remain.

- Once equilibrium is attained no net current will flow through it. because diffusion current becomes equal to drift current, hence diode is at equilibrium.
- The electric field establishes potential difference V_0 at the junction, which act as a barrier for majority charge carriers to cross the junction.
- The oppositely charged ions produces electric field E at the junction directed as shown in fig.

DRIFT & DIFFUSION CURRENT:

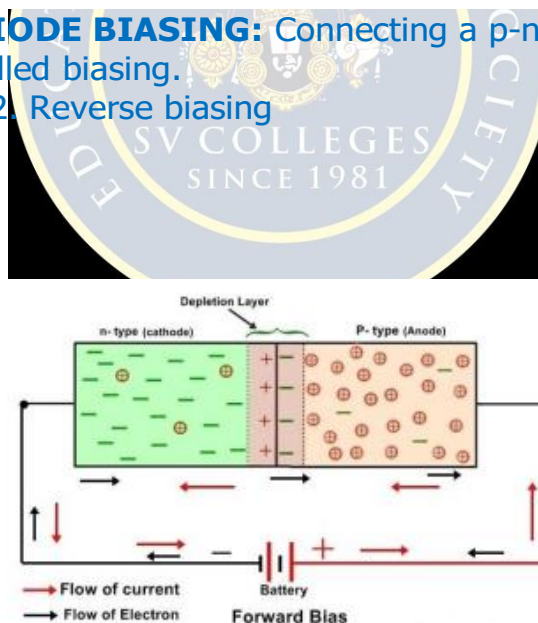
Diffusion current is the movement caused by variation in the carrier concentration. Drift current is the movement caused by electric fields. Direction of the diffusion current depends on the slope of the carrier concentration. Direction of the drift current is always in the direction of the electric field.



P-N JUNCTION DIODE BIASING: Connecting a p-n junction to an external DC voltage source is called biasing.

1. Forward biasing
2. Reverse biasing

1. Forward biasing:

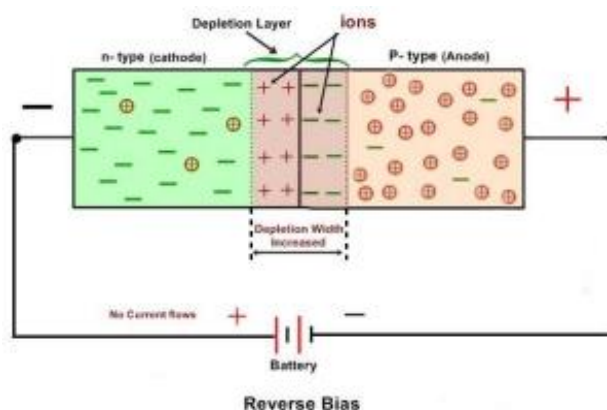


- The positive terminal of battery is connected to the P side (anode) and the negative terminal of battery is connected to the N side (cathode) of a diode, this is called forward bias.
- The holes in the p-type region and the electrons in the n-type region are pushed toward the junction and start to neutralize the depletion zone, reducing its width.
- The positive potential applied to the p-type material repels the holes, while the negative potential applied to the n-type material repels the electrons. The change in potential between the p side and the n side decreases.
- The applied forward potential (V_F) establishes the electric field which acts

against the field due to potential barrier. Therefore, the resultant field is weakened and the barrier height is reduced at the junction as shown in fig.

- Since the potential barrier voltage is very small, a small forward voltage (VF) is sufficient to completely eliminate the barrier. Once the potential barrier is eliminated by the forward voltage, junction resistance (RF) becomes almost zero and a low resistance path is established for the entire circuit. Therefore, current flows in the circuit. This is called forward current (IF).
- When a junction diode is Forward Biased the thickness of the depletion region reduces and the diode acts like a short circuit allowing full circuit current to flow.

2. Reverse biasing



- The positive terminal of battery is connected to the N side (cathode) and the negative terminal of battery is connected to the P side (anode) of a diode this is called reverse bias.
- The 'holes' in the p-type material are pulled away from the junction, leaving behind charged ions and causing the width of the depletion region to increase. Likewise, because the n-type region is connected to the positive terminal, the electrons will also be pulled away from the junction, with similar effect.
- This increases the voltage barrier causing a high resistance to the flow of charge carriers, thus allowing minimal electric current to cross the p-n junction. The increase in resistance of the p-n junction results in the junction behaving as an insulator.
- The increased potential barrier prevents the flow of charge carriers across the junction. Thus, a high resistance path (RR) is established for the entire circuit and hence current does not flow. But in practice very little current flows due to minority charge carriers. The current is called reverse saturation current (IS).
- When a junction diode is Reverse Biased the thickness of the depletion region increases and the diode acts like an open circuit blocking any current flow, (only a very small leakage current will flow).

VI CHARACTERISTICS OF PN JUNCTION DIODE:

Forward Bias:

- When the junction is forward biased, practically no current flows until the barrier voltage is overcome.
- With further increase in voltage, current starts flowing & curve has a linear rise like an ordinary conductor.
- The voltage at which current starts flowing through the diode is known as cut-in voltage.
- It is the minimum value of voltage required to overcome potential barrier across

the junction.

- This voltage is 0.3 for Germanium and 0.7 V for silicon.
- The current in forward bias mode is large (in mA) due to the movement of majority charge carriers across the junction

Reverse Bias:

- When a junction is reversed biased, junction resistance increases with the increase in reverse biased voltage (potential barrier) & very small current flow through it due to minority charge carriers.
- As reverse voltage increases, the minority charge carriers acquire sufficient kinetic energy.
- These accelerated charge carriers cross the junction and ionize the semiconductor atoms in the depletion region leading to generation of free electrons and holes.
- These electrons and holes in turn get accelerated and cause further ionization of atoms in P and N regions.
- This process is called Avalanche breakdown and leads to sharp increase in current at a reverse bias voltage called breakdown voltage.
- Thus, breakdown voltage is the applied reverse voltage at which large current flows through the diode.
- In reverse bias condition the current through diode is very small in the range of μA due to movement of minority charge carriers

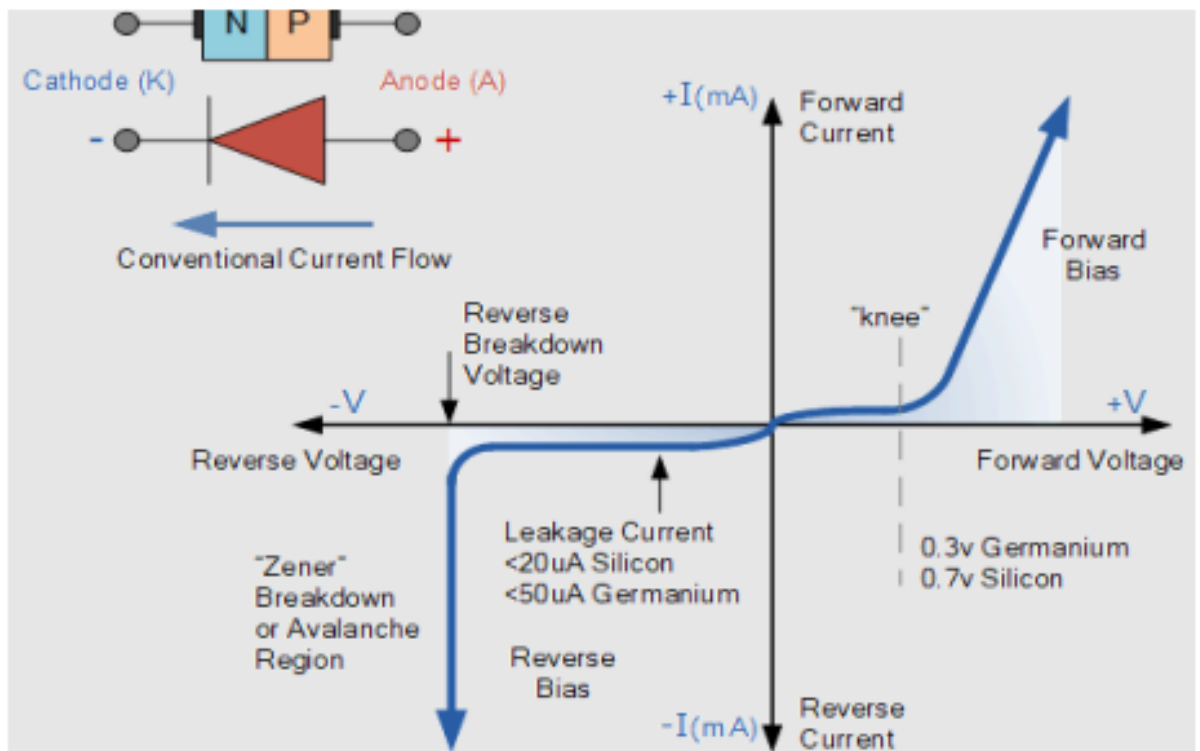


Fig. VI Characteristics of PN Junction diode

Advantages of P-N Junction Diode:

Low cost- PN junction diodes can be made easily, making them very cost-effective compared to other types.

High efficiency- PN junction diodes have high efficiency and low forward voltage drops making them perfect for power conversion applications.

Fast switching speed- PN junction diodes are fast in switching speed, making them ideal for high-frequency applications.

Wide operating temperature range- PN junction diodes are able to operate at a wide temperature range making them suitable for harsh environments.

High reliability- PN junction diodes can be used in critical applications because they are reliable and long-lasting.

Small size- PN junction diodes can be purchased in small packages making them ideal for compact designs.

Versatility- PN junction diodes are versatile components that can be used for a wide range of purposes, such as rectification, voltage regulation and protection

Disadvantages of p-n junction diode

- **Reverse voltage is limited:** PN junction diodes are not suitable for high-voltage applications due to their low reverse voltage rating.
- **Non-linear characteristics in current-voltage:** The current-voltage properties of PN junction diodes can cause distortion in applications like rectification.
- **High reverse leakage current:** PN Junction diodes have a large reverse leakage current. This can lead to power losses in reverse-biased situations.
- **Low frequency response:** PN junction diodes are not suitable for high-frequency applications due to their limited frequency response.
- **Temperature Sensitivity:** PN junction diodes can be affected by temperature. This can cause their forward voltage drop or reverse leakage current to decrease.
- **Reverse recovery time is limited:** PN junction diodes only have a short reverse recovery time. This can lead to high switching losses in fast-switching situations.
- **Limitation on voltage blocking ability:** PN junction diodes are limited in their voltage blocking capabilities, which limits their use for high-voltage applications.

Applications of P-N Junction Diode:

P-N Junction Diodes are used in the following Applications, such as

- in clamping circuits for DC restoration.
- in clipping circuits for wave shaping.
- in voltage multipliers.
- as switch in digital logic circuits used in computers.
- in demodulation circuits.
- Laser diodes are used in optical communications.
- Light Emitting Diodes (LEDs) are used in digital displays.
- in voltage regulators.
- Reverse Current Protection Circuits
- In Logic Gates
- In Voltage Multipliers etc.

1.3 ZENER EFFECT:

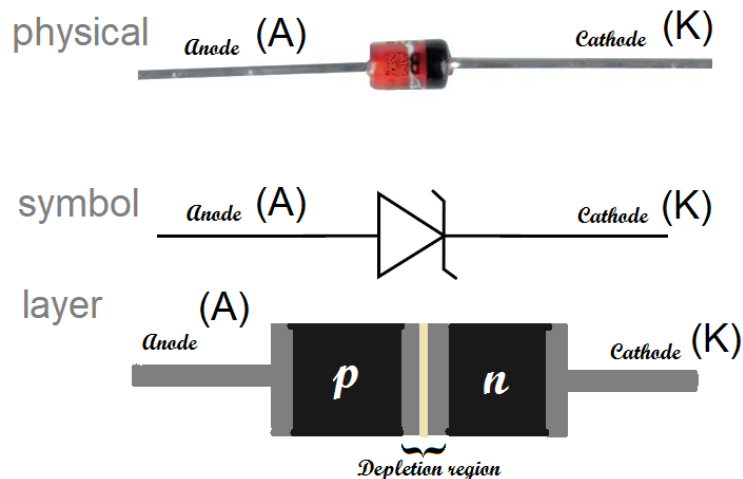
The Zener effect is a type of electrical breakdown that occurs in a reverse-biased PN junction when the electric field enables tunnelling of electrons from the valence to the conduction band of a semiconductor, leading to a large number of free minority carriers which suddenly increase the reverse current. The Zener effect is best-known for its use in the appropriately-named Zener diode.

1.4 ZENER DIODE AND ITS CHARACTERISTICS:

A Zener diode is a special type of device designed to operate in the Zener breakdown region. Therefore, Zener diodes allow more electric current than the

normal p-n junction diodes. A Zener diode is a p-n junction semiconductor device designed to operate in the reverse breakdown region. Zener diode is heavily doped than the normal p-n junction diode. Hence, it has very thin depletion region. Zener diode allows electric current in forward direction like a normal diode but also allows electric current in the reverse direction if the applied reverse voltage is greater than the Zener voltage. Zener diode is always connected in reverse direction because it is specifically designed to work in reverse direction.

Zener Diode



Zener diode definition

A Zener diode is a p-n junction semiconductor device designed to operate in the reverse breakdown region. The breakdown voltage of a zener diode is carefully set by controlling the doping level during manufacture. The name zener diode was named after the American physicist Clarence Melvin Zener who discovered the zener effect. Zener diodes are the basic building blocks of electronic circuits. They are widely used in all kinds of electronic equipment's. Zener diodes are mainly used to protect electronic circuits from over voltage.

Breakdown in Zener diode

There are two types of reverse breakdown regions in a zener diode:

- avalanche breakdown and
- Zener breakdown.

Avalanche breakdown

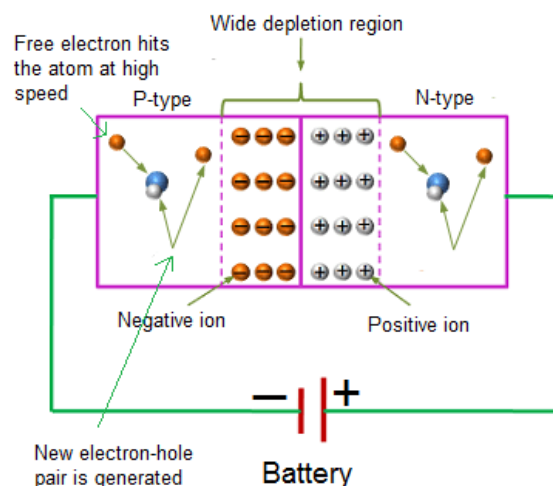


Fig: avalanche breakdown

- The avalanche breakdown occurs in both normal diodes and Zener diodes at high reverse voltage. When high reverse voltage is applied to the p-n junction diode, the free electrons (minority carriers) gain large amount of energy and accelerated to greater velocities.
- The free electrons moving at high speed will collide with the atoms and knock off more electrons. These electrons are again accelerated and collide with other atoms. Because of this continuous collision with the atoms, a large number of free electrons are generated. As a result, electric current in the diode increases rapidly. This sudden increase in electric current may permanently destroy the normal diode. However, avalanche diodes may not be destroyed because they are carefully designed to operate in avalanche breakdown region. Avalanche breakdown occurs in Zener diodes with Zener voltage (V_z) greater than 6V.

Zener breakdown

The Zener breakdown occurs in heavily doped p-n junction diodes because of their narrow depletion region. When reverse biased voltage applied to the diode is increased, the narrow depletion region generates strong electric field. When reverse biased voltage applied to the diode reaches close to zener voltage, the electric field in the depletion region is strong enough to pull electrons from their valence band. The valence electrons which gain sufficient energy from the strong electric field of depletion region will break bonding with the parent atom. The valence electrons which break bonding with parent atom will become free electrons. These free electrons carry electric current from one place to another place. At Zener breakdown region, a small increase in voltage will rapidly increase the electric current.

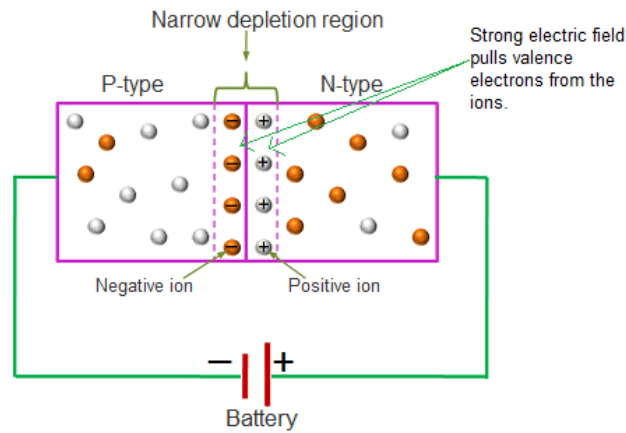


Fig: Zener breakdown

- Zener breakdown occurs at low reverse voltage whereas avalanche breakdown occurs at high reverse voltage.
- Zener breakdown occurs in Zener diodes because they have very thin depletion region.
- Breakdown region is the normal operating region for a Zener diode.
- Zener breakdown occurs in Zener diodes with Zener voltage (V_z) less than 6V.

VI Characteristics of Zener diode

The VI characteristics of a Zener diode is shown in the below figure. When forward biased voltage is applied to the Zener diode, it works like a normal diode. However, when reverse biased voltage is applied to the Zener diode, it works in different manner i.e. In reverse voltage conditions a small amount of current flows through the Zener diode. This current is because of the electrons which are thermally generated in the Zener diode. As we keep increasing the reverse voltage at any particular value of reverse voltage the reverse current increases suddenly at the breakdown point this voltage is called Zener Voltage and is represented as V_z .

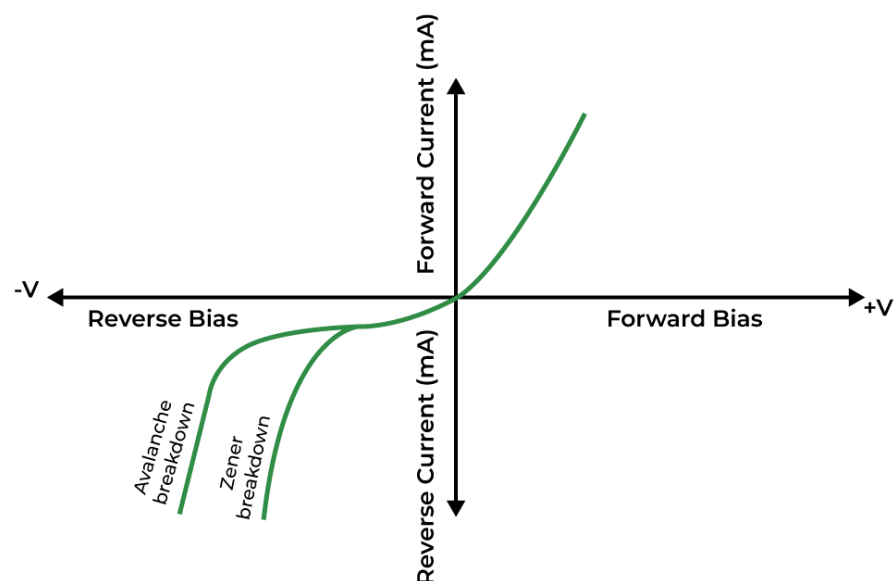


Fig: V-I Characteristics of a Zener Diode

Advantages of Zener diode

- Power dissipation capacity is very high
- High accuracy
- Small size
- Low cost

Disadvantages of Zener diode

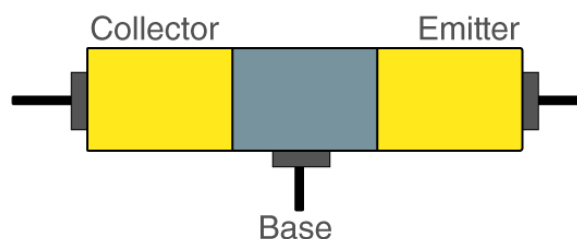
- Power rating is limited
- Can't handle high power applications
- Not suitable to use in power supplies

Applications of Zener diode

- It is normally used as voltage reference
- Zener diodes are used in voltage stabilizers or shunt regulators.
- Zener diodes are used in switching operations
- Zener diodes are used in clipping and clamping circuits.
- Zener diodes are used in various protection circuits
- Zener diode as Voltage Regulator
- Zener Diode in Over-Voltage Protection
- Zener Diode in Clipping Circuits

1.5 BIPOLAR JUNCTION TRANSISTOR (BJT)**What is a Transistor?**

A transistor is an electronic component that is used in circuits to either amplify or switch electrical signals or power, allowing it to be used in a wide array of electronic devices. A transistor consists of two PN diodes connected back-to-back. It has three terminals namely emitter, base and collector. The basic idea behind a transistor is that it lets you control the flow of current through one channel by varying the intensity of a much smaller current that's flowing through a second channel. A transistor is a semiconductor device with three terminals, capable of amplification and rectification.

Terminals of Transistor

- a transistor is composed of three terminals: emitter, collector, and base
- A transistor is a semiconductor device that contains three regions separated by two distinct PN junctions. The two junctions are EB junction and CB junctions.
- The central region is called base.
- The two outer regions are called emitter and collector.
- There are two types of charge carriers, electrons and holes hence transistor is called bipolar transistor.
- Transistor can be viewed as two PN junction diodes arranged back-to-back

- with base being common to both the diodes.
- As soon as the two junctions are formed, majority charge carriers diffuse and form two depletion layers. Depletion layer is narrow at EB junction and wide at CB junction

FUNCTIONS OF THREE REGIONS IN TRANSISTOR:

Emitter: One side of a transistor, which is of moderate size but heavily doped is known as an emitter. Its function is to supply majority charge carriers to the base region.

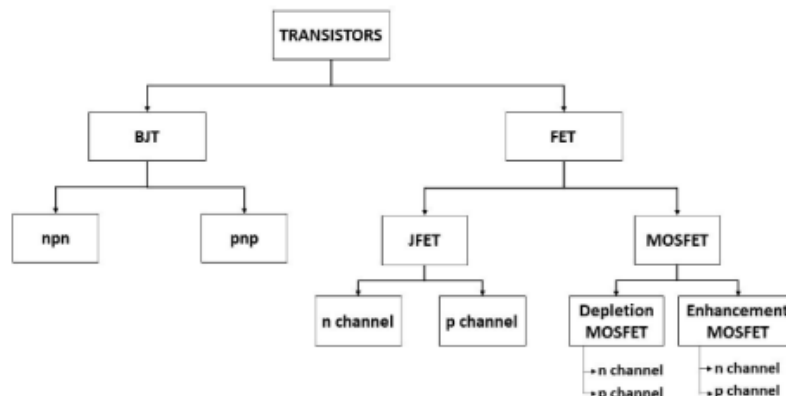
Base: The middle region of a transistor, which is very thin and lightly doped is called as Base. It controls the flow of charge carriers from emitter to collector. Therefore, it acts as a gate between emitter and collector.

Collector: The other side of Transistor, which is moderately doped but having thickness slightly more than that of the emitter (Large size) is called the collector. Its function is to collect majority charge carriers from the base region.

Biasing of transistor

Biasing is the process of providing DC voltage which helps in the functioning of the circuit. A transistor is biased in order to make the emitter base junction forward biased and collector base junction reverse biased, so that it maintains in active region, to work as an amplifier. The proper flow of zero signal collector current and the maintenance of proper collector emitter voltage during the passage of signal is known as Transistor Biasing. The circuit which provides transistor biasing is called as Biasing Circuit.

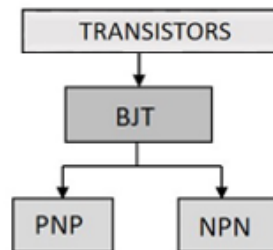
Classification Of Transistors



Bipolar Junction Transistor	Field Effect Transistor
It is a bipolar device	It is a unipolar device
It is a current-driven device	It is a voltage-driven device
Low input impedance	High input impedance
Low noise level	High noise level
Less thermal stability	Better thermal stability

Bipolar Junction Transistor (BJT)

Bipolar Junction Transistor (BJT) is one type of switching device which consists of 3 layer (semiconductors), 2 junctions, 3 terminals and bipolar. The transistor is an electronic device which is capable of achieving amplification of the weak signal from low resistance circuit to high resistance circuit. The bipolar junction transistor or BJT is a three-terminal semiconductor device that can act as a conductor or insulator based on the applied input signal. And due to this property, the transistor can be used as a switch in digital electronics or as an amplifier in analog electronics. Nowadays, the field-effect transistors are widely used in electronics applications but still, these BJTs are quite extensively used and anyone who is interested in electronics should have some basic knowledge of this device.



CONSTRUCTION OF BIPOLAR JUNCTION TRANSISTOR

The BJT is formed by three layers of semiconductor materials, if it is a PNP transistor, it will have two P-type regions and one N-type region, likewise, if it is an NPN transistor, it will have two N-type regions and one P-type region. The two outer layers are where the collector and emitter terminals are fixed and the base terminal is fixed at the center layer. The construction can simply be explained with a two-diode analogy for transistor as shown in the above image, if you want to learn more about diodes, you can consider reading his article. Consider the two diodes connected with each other using the cathode, then the meeting point can be extended to form the base terminal and the two anodes end acts as the collector and emitter of a PNP transistor. Similarly, if you connect the anode ends of the Diode then the meeting point of the anodes can be extended to for the base terminal and the two cathode ends act as the collector and emitter of the NPN transistor.

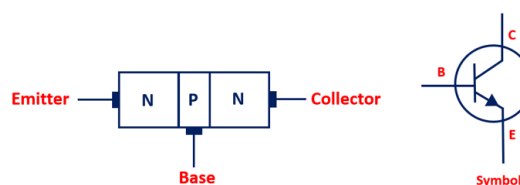


Fig. Basic Structure and Symbol of Bipolar Junction Transistor (NPN Transistor)

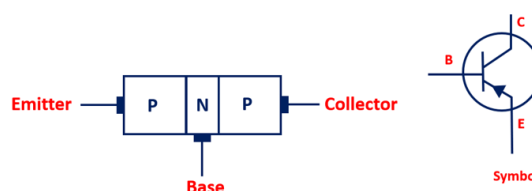
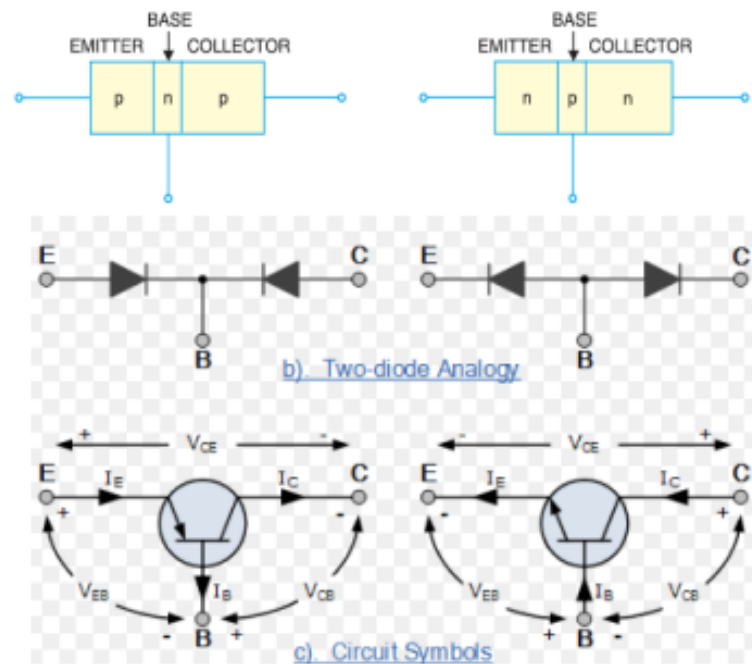


Fig. Basic Structure and Symbol of Bipolar Junction Transistor (PNP Transistor)



TERMINALS OF BIPOLAR JUNCTION TRANSISTORS :

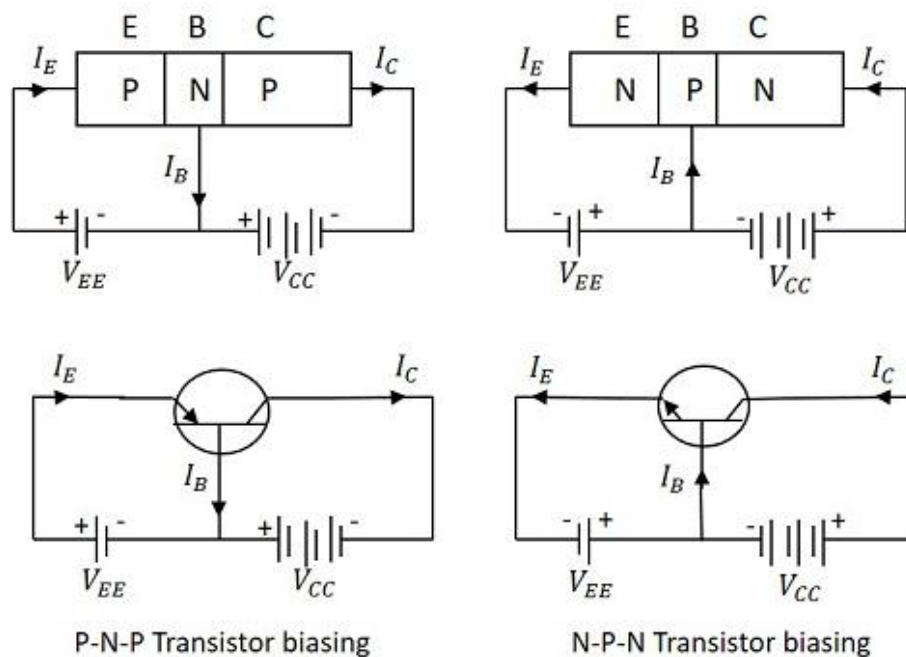
The transistor has three terminals, namely; emitter, base and collector. The base is much thinner than the emitter while collector is wider than both as shown in figure. sake of convenience, it is customary to show emitter and collector to be of equal size. The emitter is heavily doped so that it can inject a large number of charge carriers (electrons or holes) into the base. The base is lightly doped and very thin; it passes most of the emitter injected charge carriers to the collector. The collector is moderately doped.

- **Emitter** – It supplies charge carriers. It is highly doped so that it can inject a large number of charge carriers into the base. Emitter is always greater than base.
- **Base** – Base is middle layer in BJT which is thin compared to emitter and collector. Base is very lightly doped.
- **Collector** – It collects charge carriers. It's doped between emitter and base means moderately doped, but it is always greater than emitter and base in size.

BJT (NPN and PNP) TRANSISTOR BIASING:

As we know that a transistor is a combination of two diodes, we have two junctions here. As one junction is between the emitter and base, that is called as Emitter-Base junction and likewise, the other is Collector-Base junction.

Biasing is controlling the operation of the circuit by providing power supply. The function of both the PN junctions is controlled by providing bias to the circuit through some dc supply. The figure below shows how a transistor is biased.



By having a look at the above figure, it is understood that

- The N-type material is provided negative supply and P-type material is given positive supply to make the circuit **Forward bias**.
- The N-type material is provided positive supply and P-type material is given negative supply to make the circuit **Reverse bias**.
- By applying the power, the emitter base junction is always forward biased as the emitter resistance is very small. The collector base junction is reverse biased and its resistance is a bit higher. A small forward bias is sufficient at the emitter junction whereas a high reverse bias has to be applied at the collector junction.
- The direction of current indicated in the circuits above, also called as the Conventional Current, is the movement of hole current which is opposite to the electron current.

WORKING OF BJT (NPN) TRANSISTOR:

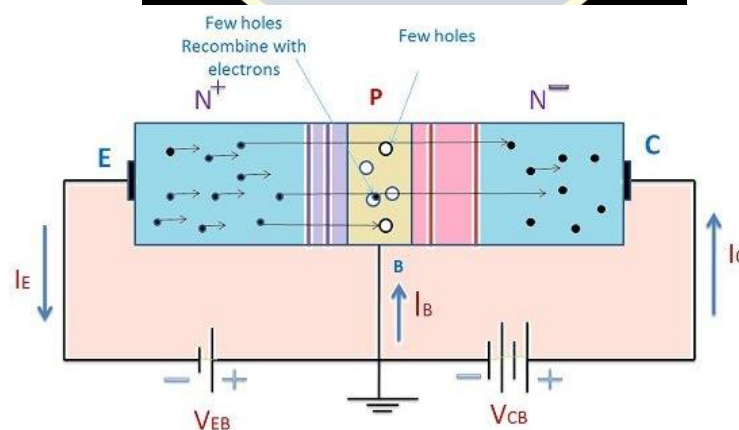


Figure shows the npn transistor with forward bias to emitter base junction and reverse bias to collector-base junction. The forward bias causes the electrons in the n-type emitter to flow towards the base. This constitutes the emitter current I_E . As these electrons flow through the p-type base, they tend to combine with holes. As the base is lightly doped and very thin, therefore, only a few electrons (less than 5%) combine with holes to constitute base current I_B . The remainder (more than 95%) cross over into the collector region to constitute collector current I_C . In this way, almost the entire emitter

current flows in the collector circuit. It is clear that emitter current is the sum of collector and base currents i.e., $I_E = I_B + I_C$.

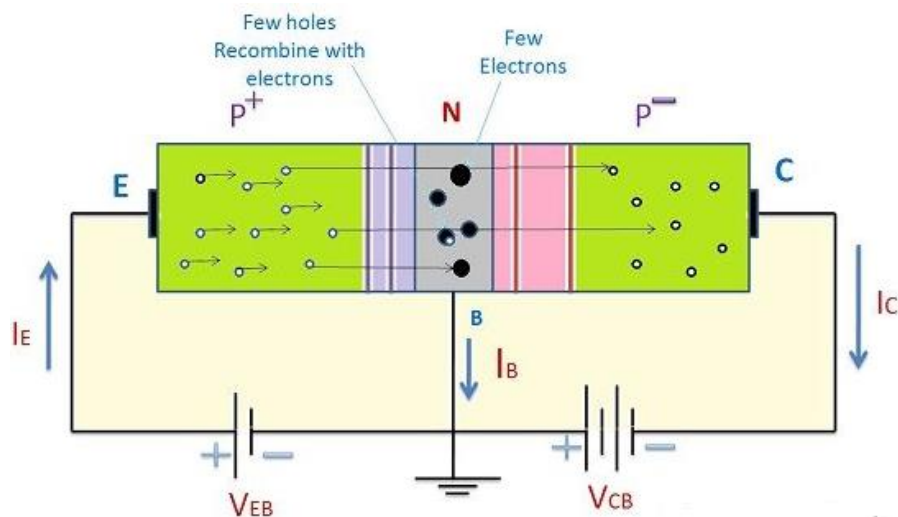
WORKING OF BJT (PNP) TRANSISTOR

Figure shows the pnp transistor with forward bias to emitter base junction and reverse bias to collector-base junction. The emitter-base junction is connected in forward biased due to which the emitter pushes the holes in the base region. These holes constitute the emitter current I_E . When these electrons move into the N-type semiconductor material or base, they combined with the electrons. The base of the transistor is thin and very lightly doped. Hence only a few holes combined with the electrons and the remaining are moved towards the collector space charge layer. Hence develops the base current I_B .

The collector base region is connected in reverse biased. The holes which collect around the depletion region when coming under the impact of negative polarity collected or attracted by the collector. This develops the collector current I_C . The complete emitter current flows through the collector circuit. It is clear that emitter current is the sum of collector and base currents i.e.,

$I_E = I_B + I_C$. and the collector current is the difference of emitter and base currents i.e.,

$$I_C = I_E - I_B.$$

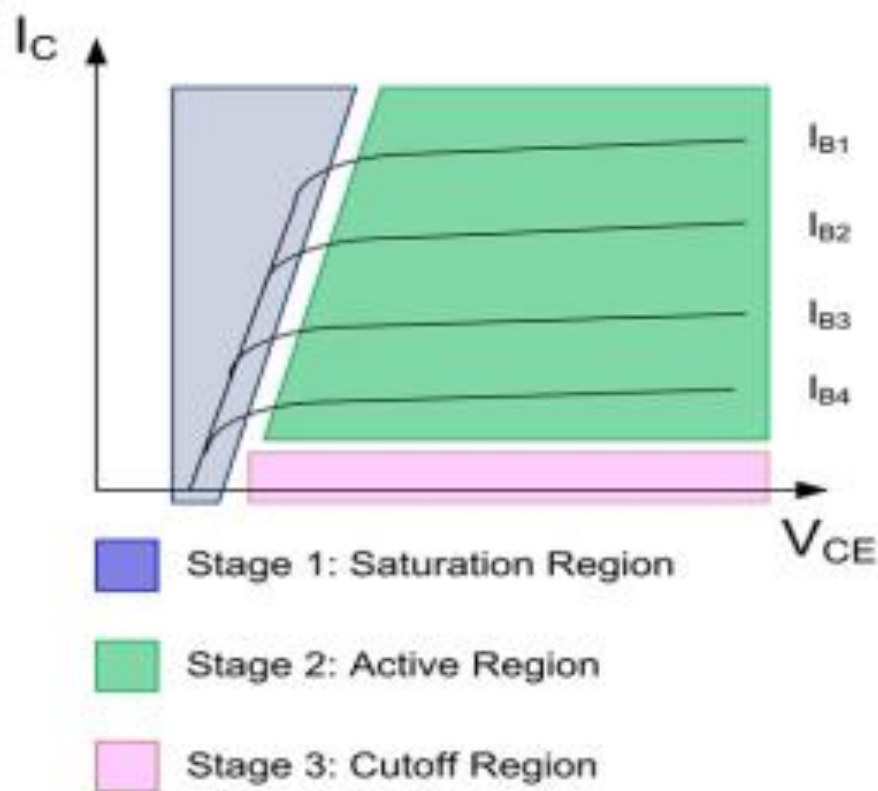


BJT TRANSISTOR (NPN/PNP) OPERATION MODES

The transistor can be operated in three modes:

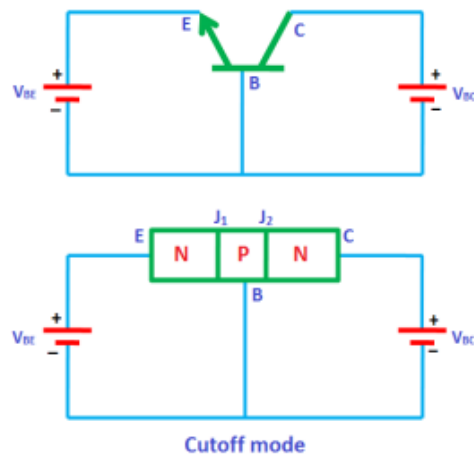
- Cut-off mode
- Saturation mode
- Active mode and another region is
- Reverse Active Mode Operation

Modes	EBJ	CBJ	Application
Cutoff	Reverse	Reverse	Switching application in digital circuits
Saturation	Forward	Forward	
Active	Forward	Reverse	Amplifier
Reverse active	Reverse	Forward	Performance degradation

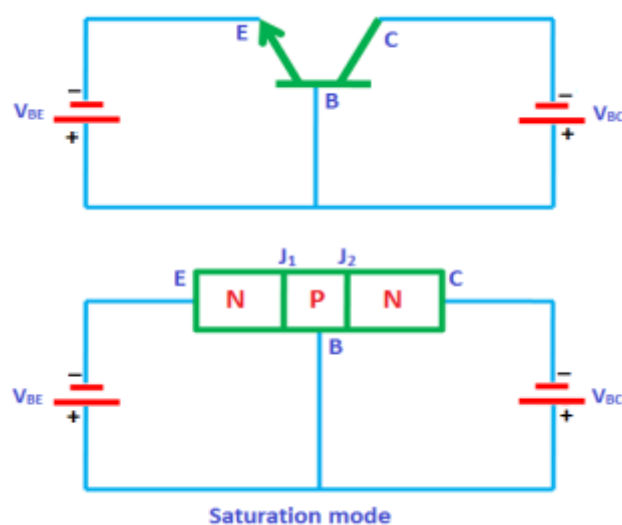


In order to operate transistor in one of these regions, we have to supply dc voltage to the npn or pnp transistor. Based on the polarity of the applied dc voltage, the transistor operates in any one of these regions. Applying dc voltage to the transistor is nothing but the biasing of transistor.

CUTOFF MODE: In the cutoff mode, both the junctions of the transistor (emitter to base and collector to base) are reverse biased. In other words, if we assume two p-n junctions as two p-n junction diodes, both the diodes are reverse biased in cutoff mode. We know that in reverse bias condition, no current flows through the device. Hence, no current flows through the transistor. Therefore, the transistor is in off state and acts like an open switch. The cutoff mode of the transistor is used in switching operation for switch OFF application.

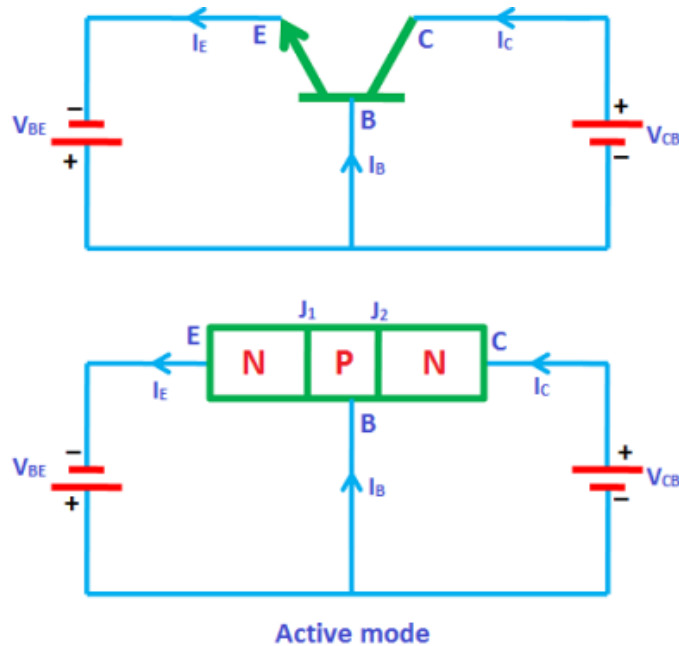


SATURATION MODE: In the saturation mode, both the junctions of the transistor (emitter to base and collector to base) are forward biased. In other words, if we assume two p-n junctions as two p-n junction diodes, both the diodes are forward biased in saturation mode. We know that in forward bias condition, current flows through the device. Hence, electric current flows through the transistor. In saturation mode, free electrons (charge carriers) flow from emitter to base as well as from collector to base. As a result, a huge current will flow to the base of transistor. Therefore, the transistor in saturation mode will be in on state and acts like a closed switch. The saturation mode of the transistor is used in switching operation for switch ON application. From the above discussion, we can say that by operating the transistor in saturation and cutoff region, we can use the transistor as an ON/OFF switch.



ACTIVE MODE: In the active mode, one junction (emitter to base) is forward biased and another junction (collector to base) is reverse biased. In other words, if we assume two p-n junctions as two p-n junction diodes, one diode will be forward biased and another diode will be reverse biased. The active mode of operation is used for the amplification of current.

From the above discussion, we can say that the transistor works as an ON/OFF switch in saturation and cutoff modes whereas it works as an amplifier of current in active mode.



Reverse Active Mode:

The reverse active mode is just opposite to the active mode because in this mode of operation the emitter-base junction is reverse biased while the collector-base junction is forward biased. As the gain is negligible in this mode of operation, therefore the transistors are never operated in reverse active mode. Because it is just opposite to the transistor action that we desire. To understand the reverse active mode with the following fig.

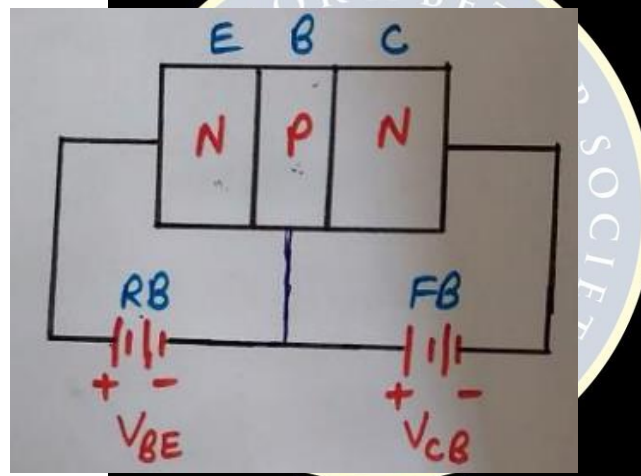


Fig. Reverse Active Mode Operation of BJT (Bipolar Junction Transistor) NPN

1.6 BJT Transistor- CB, CE, CC Configurations and Characteristics:

As mentioned earlier, when BJT is used for the amplification of the signal, it is operated in the active region. And there are different ways to configure it.

- Common Emitter (CE): has both Current and Voltage Gain.
- Common Base (CB): has Voltage Gain but no Current Gain
- Common Collector (CC): has Current Gain but no Voltage Gain.

Depending on the requirement and the application, the BJT can be configured in any of the three configurations.

Common Base (CB) Configuration:

In this (CB) configuration, for the AC signal, the base terminal is common between the input and the output. The AC input signal is applied between the emitter and the base

terminal, while the output is measured between the collector and the base terminal.

Common Emitter (CE) Configuration:

In the case of the common-emitter configuration, the emitter terminal is common between the input and the output for the AC signal. That means in this configuration the AC input signal is applied between the base and the emitter while the output is measured between the collector and the emitter.

Common Collector (CC) Configuration:

In this (CC) configuration, the base terminal is common between the input and the output. The AC input signal is applied between the base and the collector terminal, while the output is measured between the emitter and the collector terminal.

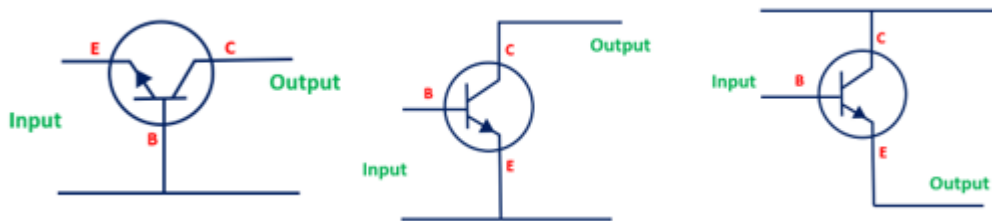


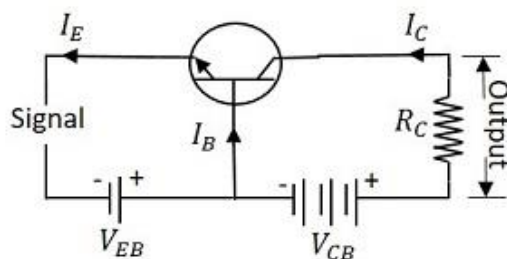
Fig. CB, CE, CC configurations.

The Common Base (CB) Configuration:

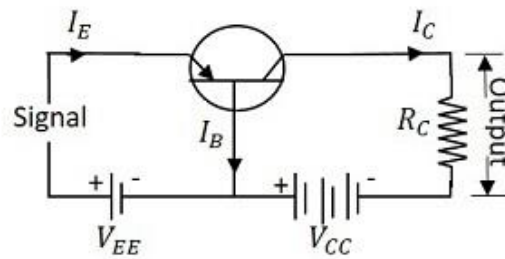
The configuration in which the base of the transistor is common between emitter and collector circuit is called a common base configuration. In common base-emitter connection, the input is connected between emitter and base while the output is taken across collector and base. The common base configuration for both NPN and PNP transistors are shown in the below figures.



Common Base Connection



Using NPN transistor

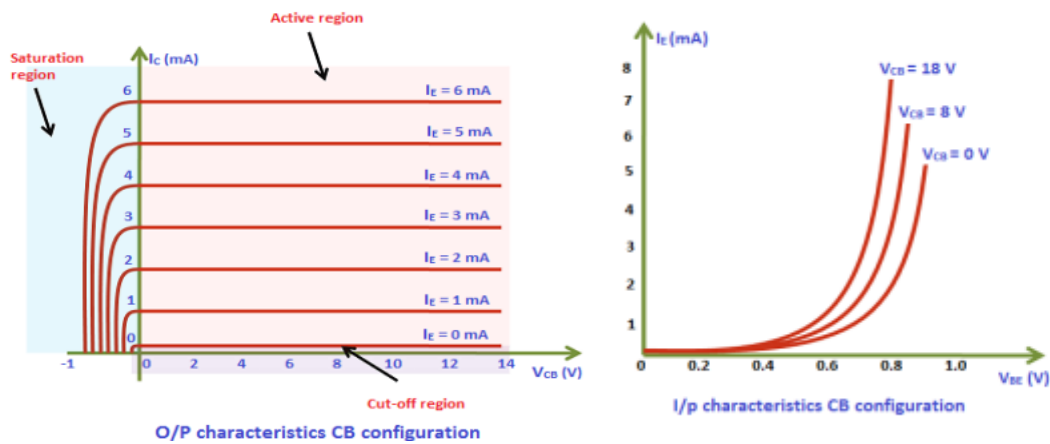


Using PNP transistor

To fully describe the behavior of a transistor with CB configuration, we need two set of characteristics: they are

- Input characteristics
- Output characteristics.

Input and Output Characteristics of Common Base Configuration:



Input characteristics: Input current (I_E) vs input voltage (V_{BE}) at constant output voltage (V_{CB}):

The input characteristics describe the relationship between input current (I_E) and the input voltage (V_{BE}). First, draw a vertical line and horizontal line. The input current or emitter current (I_E) is taken along the y-axis (vertical line) and the input voltage (V_{BE}) is taken along the x-axis (horizontal line).

To determine the input characteristics, the output voltage V_{CB} (collector-base voltage) is kept constant at zero volts and the input voltage V_{BE} is increased from zero volts to different voltage levels. For each voltage level of the input voltage (V_{BE}), the input current (I_E) is recorded. A curve is then drawn between input current I_E and input voltage V_{BE} at constant output voltage V_{CB} (0 volts). Next, the output voltage (V_{CB}) is increased from zero volts to a certain voltage level (8 volts) and kept constant at 8 volts. A curve is then drawn between input current I_E and input voltage V_{BE} at constant output voltage V_{CB} (8 volts). This is repeated for higher fixed values of the output voltage (V_{CB}).

Output characteristics: output current (I_C) vs output voltage (V_{CB}) at constant input current (I_E):

The output characteristics describe the relationship between output current (I_C) and the output voltage (V_{CB}).

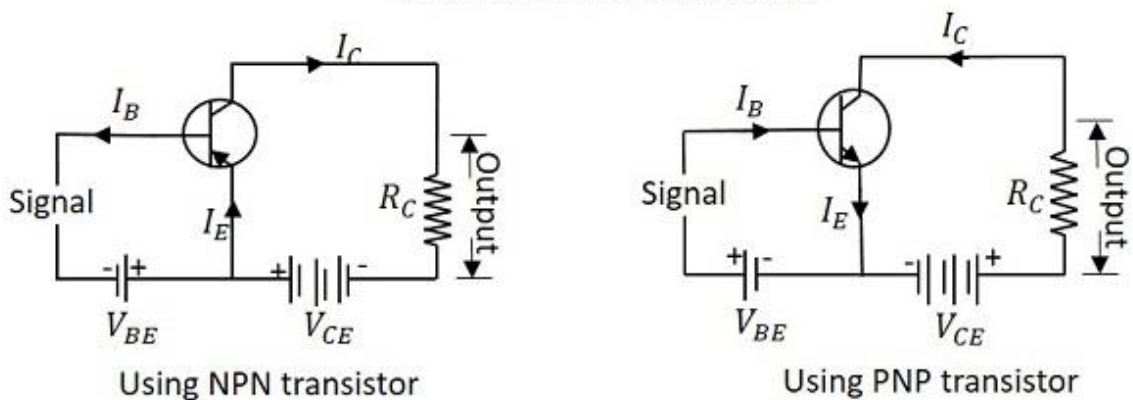
- First, draw a vertical line and a horizontal line. The output current or collector current (I_C) is taken along the y-axis (vertical line) and the output voltage (V_{CB}) is taken along the x-axis (horizontal line).
- To determine the output characteristics, the input current or emitter current I_E is kept constant at zero mA and the output voltage V_{CB} is increased from zero volts to different voltage levels.
- For each voltage level of the output voltage V_{CB} , the output current (I_C) is recorded. A curve is then drawn between output current I_C and output voltage V_{CB} at constant input current I_E (0 mA). When the emitter current or input current I_E is equal to 0 mA, the transistor operates in the cut-off region.
- Next, the input current (I_E) is increased from 0 mA to 1 mA by adjusting the input voltage V_{BE} and the input current I_E is kept constant at 1 mA. For each voltage level of the output voltage (V_{CB}), the output current (I_C) is recorded.
- A curve is then drawn between output current I_C and output voltage V_{CB} at constant input current I_E (1 mA). This region is known as the active region of a transistor. This is repeated for higher fixed values of input current I_E (i.e., 2 mA, 3 mA, 4 mA and so on).
- From the above characteristics, we can see that for a constant input current I_E , when the output voltage V_{CB} is increased, the output current I_C remains constant.
- At saturation region, both emitter-base junction J_E and collector-base junction J_C are forward biased. From the above graph, we can see that a sudden increase in

the collector current when the output voltage V_{CB} makes the collector-base junction J_C forward biased.

COMMON EMITTER (CE) CONFIGURATION:

The configuration in which the emitter is connected between the collector and base is known as a common emitter configuration. The input circuit is connected between emitter and base, and the output circuit is taken from the collector and emitter. In common emitter configuration, the emitter terminal is grounded so the common emitter configuration is also known as grounded emitter configuration. Sometimes common emitter configuration is also referred to as CE configuration, common emitter amplifier, or CE amplifier. The common emitter (CE) configuration is the most widely used transistor configuration.

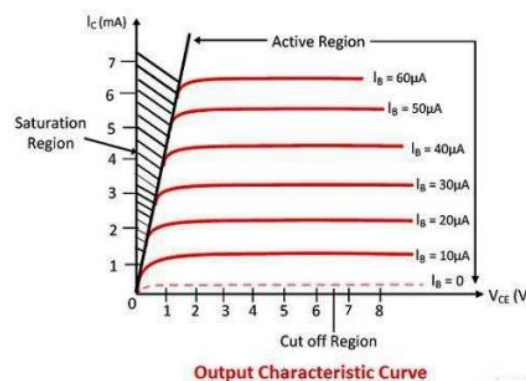
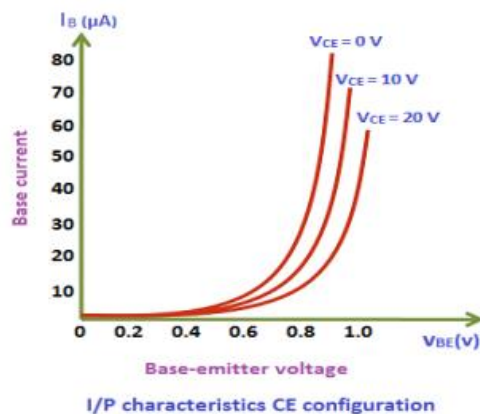
Common Emitter Connection



To fully describe the behavior of a transistor with CE configuration, we need two set of characteristics – input characteristics and output characteristics.

Input characteristics

- The input characteristics describe the relationship between input current or base current (I_B) and input voltage or base-emitter voltage (V_{BE}).
- First, draw a vertical line and a horizontal line. The vertical line represents y-axis and horizontal line represents x-axis. The input current or base current (I_B) is taken along y-axis (vertical line) and the input voltage (V_{BE}) is taken along x-axis (horizontal line).
- To determine the input characteristics, the output voltage V_{CE} is kept constant at zero volts and the input voltage V_{BE} is increased from zero volts to different voltage levels. For each voltage level of input voltage (V_{BE}), the corresponding input current (I_B) is recorded.
- A curve is then drawn between input current I_B and input voltage V_{BE} at constant output voltage V_{CE} (0 volts). Next, the output voltage (V_{CE}) is increased from zero volts to certain voltage level (10 volts) and the output voltage (V_{CE}) is kept constant at 10 volts. For each voltage level of input voltage (V_{BE}), the corresponding input current (I_B) is recorded. A curve is then drawn between input current I_B and input voltage V_{BE} at constant output voltage V_{CE} (10 volts).
- This process is repeated for higher fixed values of output voltage (V_{CE}).
- From the above characteristics, we can see that for higher fixed values of output voltage V_{CE} , the curve shifts to the right side. This is because for higher fixed values of output voltage, the cut in voltage is increased above 0.7 volts. Therefore, to overcome this cut in voltage, more input voltage V_{BE} is needed than previous case.



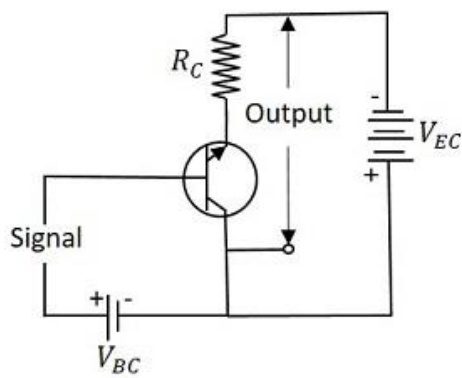
Output characteristics

- The output characteristics describe the relationship between output current (I_C) and output voltage (V_{CE}).
- First, draw a vertical line and a horizontal line. The output current or collector current (I_C) is taken along y-axis (vertical line) and the output voltage (V_{CE}) is taken along x-axis (horizontal line).
- To determine the output characteristics, the input current or base current I_B is kept constant at $0 \mu A$ and the output voltage V_{CE} is increased from zero volts to different voltage levels. For each level of output voltage, the corresponding output current (I_C) is recorded.
- A curve is then drawn between output current I_C and output voltage V_{CE} at constant input current I_B ($0 \mu A$).
- When the base current or input current $I_B = 0 \mu A$, the transistor operates in the cut-off region. In this region, both junctions are reverse biased.
- Next, the input current (I_B) is increased from $0 \mu A$ to $10 \mu A$ by adjusting the input voltage (V_{BE}). The input current (I_B) is kept constant at $10 \mu A$.
- For each voltage level of output voltage (V_{CE}), the corresponding output current (I_C) is recorded. A curve is then drawn between output current I_C and output voltage V_{CE} at constant input current I_B ($10 \mu A$). This region is known as the active region of a transistor. In this region, emitter-base junction is forward biased and the collector-base junction is reverse biased.
- These steps are repeated for higher fixed values of input current I_B (i.e. $2 \mu A$, $40 \mu A$, $60 \mu A$, $80 \mu A$ and so on). When output voltage V_{CE} is reduced to a small value ($0.2 V$), the collector-base junction becomes forward biased. This is because the output voltage V_{CE} has less effect on collector-base junction than input voltage V_{BE} . As we know that the emitter-base junction is already forward biased. Therefore, when both the junctions are forward biased, the transistor operates in the saturation region. In this region, a small increase in output voltage V_{CE} will rapidly increase the output current I_C .

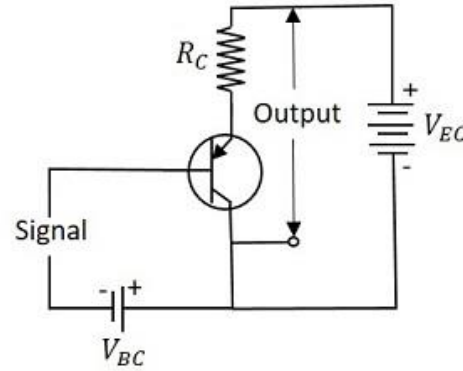
Common Collector (CC) Configuration:

The configuration in which the emitter is connected between the collector and base is known as a common emitter configuration. Thus, the emitter is common to both the input and the output circuit, and hence the name is the common emitter configuration. In common collector configuration, the collector terminal is grounded so the common collector configuration is also known as grounded collector configuration. Sometimes common collector configuration is also referred to as emitter follower, voltage follower, common collector amplifier, CC amplifier, or CC configuration. This configuration is mostly used as a voltage buffer.

Common Collector Connection



Using NPN transistor

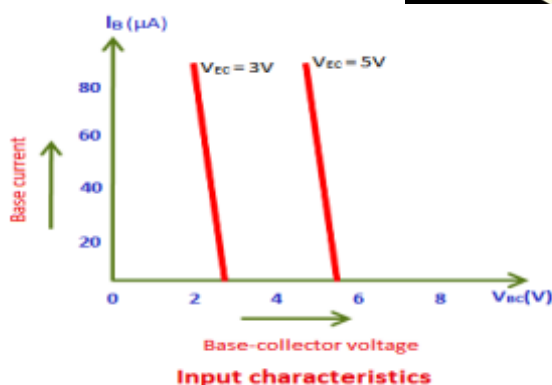


Using PNP transistor

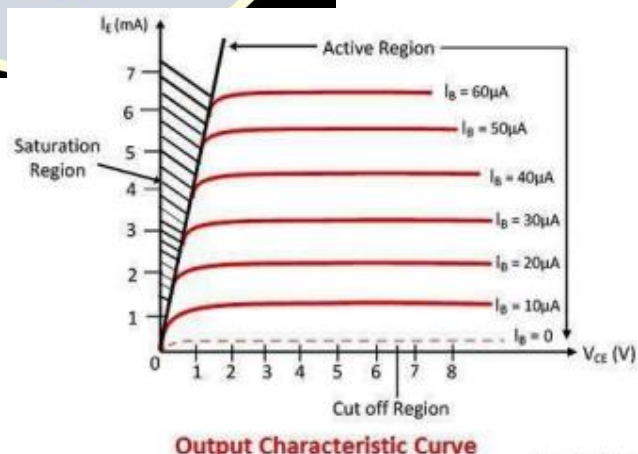
To fully describe the behavior of a transistor with CC configuration, we need two set of characteristics – input characteristics and output characteristics.

Input characteristics

- The input characteristics describe the relationship between input current or base current (I_B) and input voltage or base-collector voltage (V_{BC}).
- First, draw a vertical line and a horizontal line. The input current or base current (I_B) is taken along y-axis (vertical line) and the input voltage or base-collector voltage (V_{BC}) is taken along x-axis (horizontal line).
- To determine the input characteristics, the output voltage V_{EC} is kept constant at 3V and the input voltage V_{BC} is increased from zero volts to different voltage levels. For each level of input voltage V_{BC} , the corresponding input current I_B is noted. A curve is then drawn between input current I_B and input voltage V_{BC} at constant output voltage V_{EC} (3V). Next, the output voltage V_{EC} is increased from 3V to different voltage level, say for example 5V and then kept constant at 5V.
- For each level of input voltage V_{BC} , the corresponding input current I_B is noted. A curve is then drawn between input current I_B and input voltage V_{BC} at constant output voltage V_{EC} (5V). This process is repeated for higher fixed values of output voltage (V_{EC}).



Input characteristics



Output Characteristic Curve

Output characteristics

- The output characteristics describe the relationship between output current or emitter current (I_E) and output voltage or emitter-collector voltage (V_{EC}).
- First, draw a vertical line and a horizontal line. The output current or emitter current (I_E) is taken along y-axis (vertical line) and the output voltage or emitter-collector voltage (V_{EC}) is taken along x-axis (horizontal line).
- To determine the output characteristics, the input current I_B is kept constant at zero

micro amperes and the output voltage V_{EC} is increased from zero volts to different voltage levels.

- For each level of output voltage V_{EC} , the corresponding output current I_E is noted. A curve is then drawn between output current I_E and output voltage V_{EC} at constant input current I_B ($0 \mu A$).
- Next, the input current (I_B) is increased from $0 \mu A$ to $10 \mu A$ and then kept constant at $10 \mu A$. While increasing the input current (I_B), the output voltage (V_{EC}) is kept constant at 0 volts. For each level of output voltage (V_{EC}), the corresponding output current (I_E) is recorded. A curve is then drawn between output current I_E and output voltage V_{EC} at constant input current I_B ($10 \mu A$). This region is known as the active region of a transistor.
- This process is repeated for higher fixed values of input current I_B (I.e. $20 \mu A$, $30 \mu A$, $40 \mu A$ and so on). In common collector configuration, if the input current or base current is zero then the output current or emitter current is also zero.
- As a result, no current flows through the transistor. So the transistor will be in the cutoff region. If the base current is slightly increased then the output current or emitter current also increases.
- So the transistor falls into the active region. If the base current is heavily increased then the current flowing through the transistor also heavily increases. As a result, the transistor falls into the saturation region.

Advantages and disadvantages of BJT:

Advantages of BJT:

- They have a better voltage gain
- They have a high current density
- They have a low forward voltage
- It can be operated in low to high power application
- BJT has a large gain bandwidth
- BJT shows better performance at high frequency

Disadvantages of BJT:

- BJT has a low thermal stability
- BJT is most effective by radiation
- BJT has more noise produced
- BJT has a low switching frequency
- BJT has a very complex control
- The switching time is not very fast compared to a high alternating frequency of current and voltage

BJT Applications:

- The bipolar junction transistor (BJT) is used in logic circuits.
- The BJT is used as an oscillator.
- It is used as an amplifier.
- It is used as a multivibrator.
- For wave shaping it is used in clipping circuits.
- Used as a detector or demodulator.
- It is also used as modulator.
- Used in timer and time delay circuits.
- It is used in electronics switch.
- It is used in switching circuits.

1.7 SMALL SIGNAL CE AMPLIFIER:

When the input signal is so weak so as to produce small fluctuations in the collector current compared to its quiescent value, the amplifier is known as small signal amplifier.

The amplifier circuit that is formed using a CE configured transistor combination is called as CE amplifier.

The CE amplifier has a voltage divider bias network with a bypass capacitor CE across RE. The value of CE is chosen so that its reactance at AC frequencies of interest is very small, so that it provides an AC short circuit. The load is modelled by a single resistance RL, connected to the output of the amplifier (collector terminal) via a coupling capacitor CC. The source is modelled by a Thevenin circuit consisting of a voltage source v_s and a resistor R_s , and is connected to the input of the amplifier by a coupling capacitor CS. The coupling capacitors are chosen to provide an AC short circuit. The capacitors provide a blocking of DC (open circuit). The small signal CE amplifier circuit is shown below.

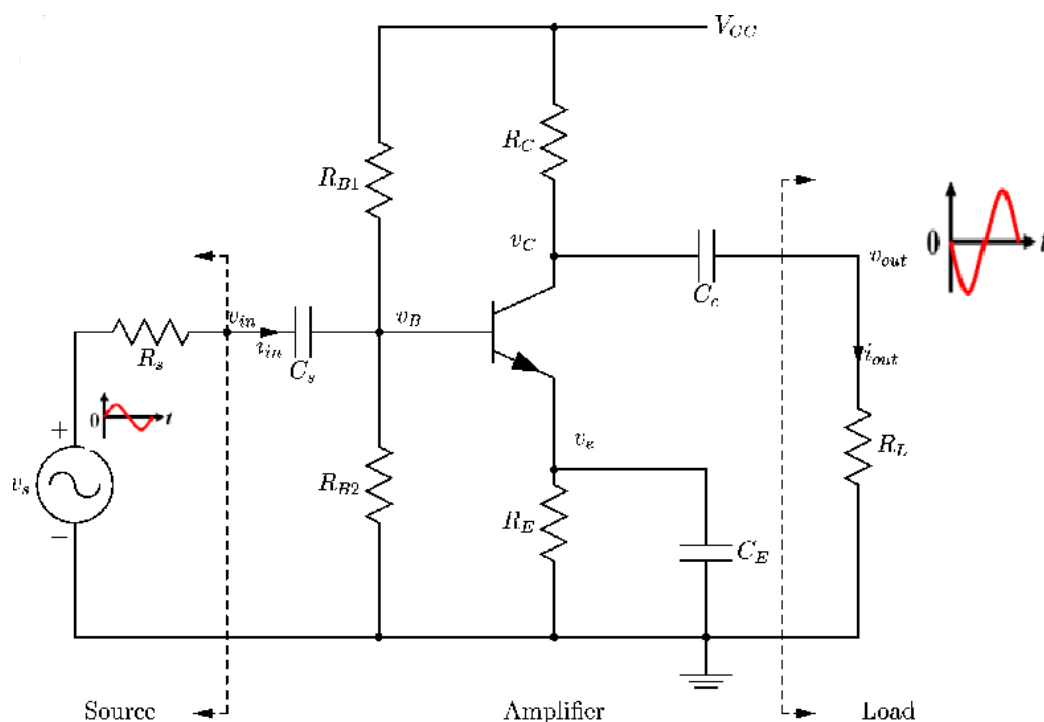


Fig: The small signal CE amplifier

Advantages of Small Signal CE amplifier:

- A common emitter amplifier is inverting and has low input impedance
- high output impedance
- high voltage gain
- high current gain

Disadvantages of Small Signal CE amplifier:

- It has a high output resistance.
- It responds poorly to high frequencies.
- It has high thermal instabilities.
- It's voltage gain is very unstable.

Applications of Small signal CE amplifier:

- used in the initial stages of public address (PA) amplifier systems,
- in stereo amplifiers,
- in tape recorder, CD players, VCRs, DVD players,
- in radio and television receivers and most widely used as a voltage amplifier.

9. Practice Quiz

1. Identify the main function of a BJT?

- A) Acts as amplifier
- B) Acts as a switch
- C) Acts as a rectifier
- D) **Both a and b**

2. In NPN BJT transistor which of the following terminal defines P-type?.

- A) Emitter
- B) Collector
- C) **Base**
- D) Drain

3. _____ junction in BJT is forward biased.

- A) Emitter
- B) Collector
- C) Base
- E) **Both a and c**

4. A BJT is a ____ electronic component.

- A) Resistor
- B) **Transistor**
- C) Capacitor
- D) Both a and c

5. In a PN junction diode, the reverse current _____ for every 10°C rise in temperature.

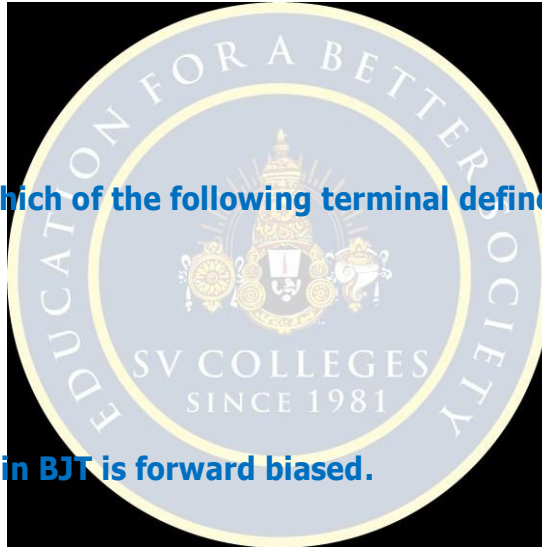
- A) Triples
- B) Multiplies by 100 times
- C) **Doubles**
- D) None of these

6. In which of the following BJT region acts like a short circuit _____.

- A) Active
- B) **Saturation**
- C) Cut-off
- D) None of these

7. A PN junction diode can conduct substantial current in the _____ direction

- A) reverse
- B) **forward**



- C) blocking
D) None of these

8. Which of the following is the application of a PN junction diode_____.

- A) Amplifier
B) Attenuator
C) Rectifier circuit
D) None of these

9. Which of the following are the application of common base type BJT_____

- A) RF amplifier
B) Microphone
C) Pre amplifier
D) All the above

10. A PNP BJT is said to be in ON when voltage of base with respect to emitter is _____

- A) Equal
B) Zero
C) Increased
D) Decreased

10. Assignments

S.No	Question	BL	CO
1	a) With neat diagram, explain the operation of p-n junction diode considering different biasing conditions. b) Illustrate V-I characteristics of p-n junction diode.	3	1
2	a) Define Zener effect. Explain the operation of Zener diode with Zener Breakdown and Avalanche Breakdown. b) Illustrate V-I characteristics of Zener diode	2	1
3	a) Illustrate the Input and output characteristics of all three configurations of a BJT and Explain. b) Explain the construction and operation of BJT.	3	1

11. Part A- Question & Answers

S.No	Question & Answers	BL	CO
1	What is doping? Ans. The process of adding impurities to an intrinsic semiconductor atom is doping.	1	1
2	What are Semiconductors? Give examples? Ans. The materials whose electrical properties lies between those of conductors and insulators are known as Semiconductors. Ex Germanium, Silicon.	1	1

3	Define the term diffusion current. Ans. A concentration gradient exists, if the number of either electrons or holes is greater in one region of a semiconductor as compared to the rest of the region. The holes and electron tend to move from region of higher concentration to the region of lower concentration. This process is called diffusion and the current produced due to this movement is diffusion current.	1	1
4	List the applications of PN junction diode. Ans. • Used as rectifier diodes in dc power supplies • Used as signal diodes in communication circuits for modulation and demodulation • Used in clipped and clamper circuits. • Used as a switch in logic circuits used in computers.	1	1
5	What is an ideal diode? Ans. An ideal diode is one which offers zero resistance when forward biased and infinite resistance when reverse biased.	1	1
6	Define knee/cut-in/threshold voltage of a PN diode. Ans. It is the forward voltage applied across the PN diode below which practically no current flows.	1	1
7	What is the effect of junction temperature on cut-in voltage of a PN diode? Ans. Cut-in voltage of a PN diode decreases as junction temperature increases.	1	1
8	What is the effect of junction temperature on forward current and reverse current of a PN diode? Ans. For the same forward voltage, the forward current of a PN diode increases and reverse saturation current increases with increase in junction temperature.	1	1
9	What is a Drift Current? Ans. 1. It is developed due to potential gradient. 2. This phenomenon is found both in metals and semiconductors.	1	1
10	List the applications of Zener diode. Ans. • used as voltage reference • used in voltage stabilizers • used in switching operations • used in clipping and clamping circuits. • used in various protection circuits • Zener diode as Voltage Regulator • used in Over-Voltage Protection • used in Clipping Circuits	1	1

12. Part B- Questions

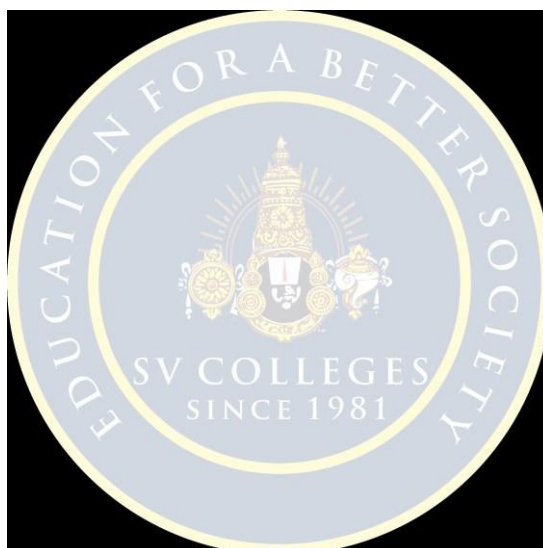
S.No	Question	BL	CO
1	a) With neat diagram, explain the operation of p-n junction diode considering different biasing conditions. b) Illustrate V-I characteristics of p-n junction diode.	3	1
2	a) Define Zener effect. Explain the operation of Zener diode with Zener Breakdown and Avalanche Breakdown. b) Illustrate V-I characteristics of Zener diode	2	1
3	With the help of a neat circuit diagram, input and output waveforms, describe the operation of small signal CE amplifier.	3	1
4	c) Illustrate the Input and output characteristics of all three configurations of a BJT and Explain. d) Explain the construction and operation of BJT.	3	1
5	a) Compare CE, CB and CC configurations of a BJT. b) Differentiate Avalanche Breakdown and Zener Breakdown	2	1

13. Supportive Online Certification Courses

1. Basic Electronics -- Web course By Dr. Pramod Agarwal, Department of Electrical Engineering, IIT Roorkee – 12 weeks
<https://nptel.ac.in/courses/117/107/117107095/#>
2. Basic Electronics -- Web course By Prof. Chitralekha Mahanta, Department of Electronics and Communication Engineering, IIT Guwahati– 12 weeks.
<https://nptel.ac.in/courses/117/103/117103063/>

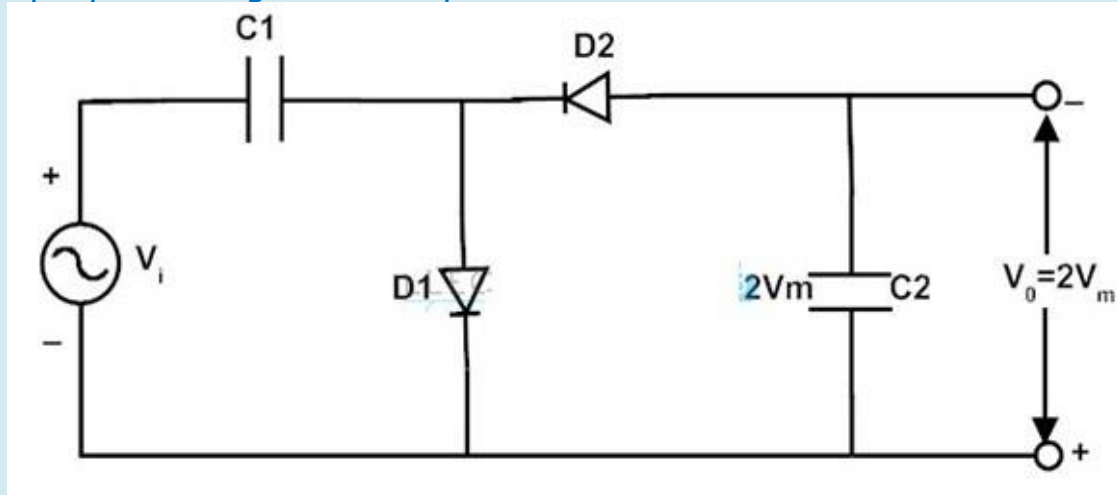
14. Real Time Applications

S.No	Application	CO
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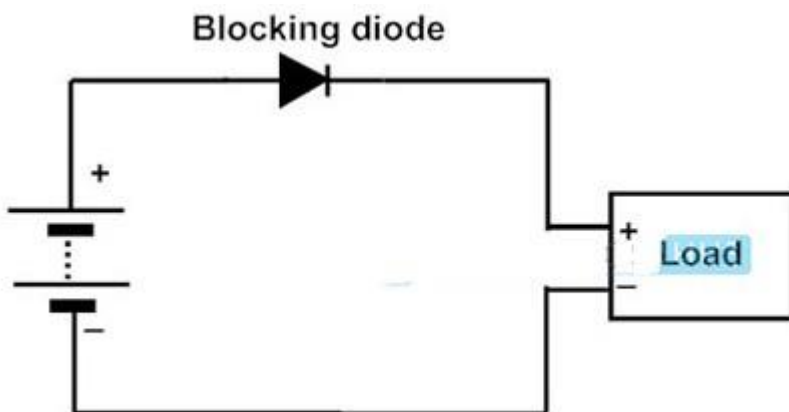
Diodes in Voltage Multiplier Circuits

Voltage multiplier consist of two or more diode rectifier circuits, which are cascaded to produce a DC output voltage equal to the multiple of the applied input voltage. These multiplier circuits are of different types like voltage doubler, tripler, quadrupler, etc. By the usage of diodes in combination with capacitors, we get the odd or even multiple of the input peak voltage at the output.



Above figure shows a half-wave voltage doubler circuit whose DC output voltage is twice that of peak input AC voltage. During the positive half-cycle of the AC input, diode D1 is forward-biased and D2 is reverse-biased. So, the capacitor C1 charges up to peak voltage V_m of the input through the diode D1. During the negative half-cycle of the AC input, D1 is reverse-biased and D2 is forward-biased. So, capacitor C2 starts charging thorough D2 and C1. Thus, the total voltage across the C2 is equal to the $2V_m$.

During next positive half-cycle, the diode D2 is reverse-biased so the capacitor C2 will discharge through the load. Likewise, by cascading the

	rectifier circuits we will get the multiple values of input voltage at the output.	
2	Diodes in Reverse Polarity Protection The reverse polarity or current protection is necessary to avoid the damage that occurs due to connecting the battery in a wrong way or reversing the polarities of the DC supply. This accidental connection of supply causes to flow a large amount current thorough the circuit components, which might result in their failure or in a worst case, their explosion. Therefore, a protective or blocking diode is connected in series with the positive side of the input to avoid the reverse connection problem.  Above figure shows the reverse current protection circuit, where diode is connected in series with the load at the positive side of the battery supply. In case of the correct polarity connection, diode gets forward-biased and load current flows through it. But, in case of wrong connection, the diode is reverse-biased and that doesn't allow any current to flow to the load. Hence, the load is protected against the reverse polarity.	4

15. Contents Beyond the Syllabus

1. Conduction of Technical Quiz

A technical quiz on Electronic materials and its properties.

2. Industrial visit to Amararaja Industries Pvt. Ltd.

Students are exposed to various electronic components and electronic boards in real time measurements.

16. Prescribed Text Books & Reference Books

Text Book

1. R.L. Boylestad and Louis Nashelsky, —Electronic Devices and Circuits,|| 9th Edition, Pearson, 2006.

2. R. P. Jain, Modern Digital Electronics, 4th Edition, Tata Mc Graw Hill, 2009.

References:

1. J. Milliman and C Halkias, –Integrated electronics||, 2nd Edition, Tata McGraw Hill, 1991.
2. Behzad Razavi, —Microelectronics||, Second edition, Wiley, 2013.
3. Adel S. Sedra and Kenneth C. Smith, —Microelectronic Circuits – Theory and Applications||, 6th Edition, Oxford Press, 2013

17. Mini Project Suggestion

1. Water level indicator

Students were asked to complete a project that involved the design and assembly of a circuit that will sense different levels of water inside a storage tank, using indicator lights. The design was expected to trigger an alarm once the water level reached full capacity. Students were also asked to prepare an excel spread sheet that calculates volume of water at different levels inside the storage tank.

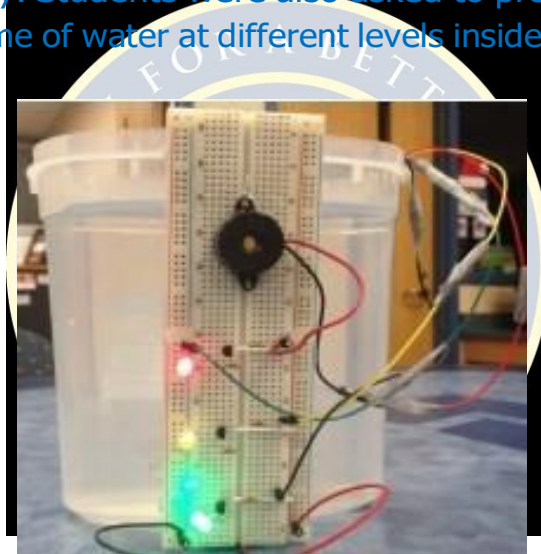


Fig. 1.12 Hardware connections of water level indicator

Sensor developed to measure levels of water inside a tank. At level 3, which indicates the tank is full, the attached buzzer alarm goes off simultaneously as the red indicator light comes on

2. Touch Switch Circuit For Doorbell Alarm Without IC

When finger is placed on the touch metal, then the Alarm is switched ON and when the finger is removed from the touch plate then the Alarm is Automatically Switch OFF. This is very simple and easy because only three transistors and some other components are used, No IC required for this Project.

Using this circuit you can change the Old Experience of Doorbell Switching method and can create a new high-tech Method with Automatic Touch Doorbell Alarm Switch System.

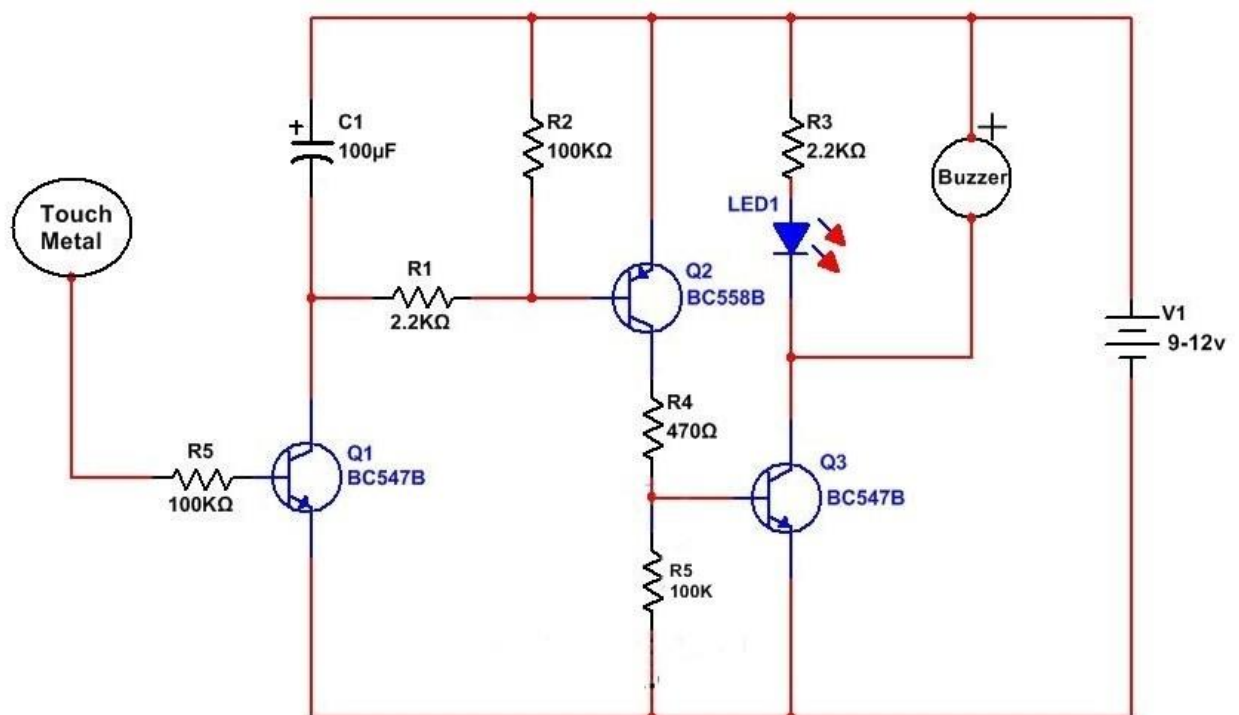


Fig. 1.13 Hardware connections for doorbell alarm

